Impact of Occupational Licensing on Interstate Migration of Workers

Arvind Sharma[*]

November 27, 2022

Labor mobility offers workers the opportunity to improve their personal and financial situations, and plays a large role in affecting unemployment and underemployment across the country. There are several factors that limit a worker's mobility, including family ties and potential inertia as well as external factors arising from regulation. One external factor that has gained prominence in recent decades is the use of (state-specific) occupational licensing requirements. Combining data from the Current Population Survey with information from the Annual Social and Economic Supplement for all occupations and states over 5 years, I study the impact of occupational licensing on a worker's interstate mobility. Using k-means algorithms to account for potential reporting issues and running a set of fixed effects models, I find little evidence that occupational licensing limits interstate mobility. Thus, the concern that occupational licensing contributes to mismatches in the labor market seems unsubstantiated.

Keywords: occupation, licensing, migration, workers

JEL Codes: J61, J62, D0

[*]Visiting Assistant Professor, Woods College of Advancing Studies, Boston College. arvind.sharma@bc.edu

1 Introduction

1.1 Motivation

There are two strong motivations to study occupational licensing (OL).

First, about one-quarter of the U.S. workforce needed a license to work in their occupation in 2019 as per National Conference of State Legislature (2019). Not only is this large in magnitude but occupational licensing already impacts more people than both union memberships and federal minimum wage coverage in US, which affected 10.3% and 1.8% of the employed Americans in 2019 (Shierholz 2020). In fact, as of July 2015, there were at least 60 occupations that were licensed in every state and over 1,100 occupations that were licensed in at least one state, as per Treasury Office of Economic Policy, Economic Advisers, and Labor (2015).

Furthermore, occupational licensing as a labor market institution has been steadily gaining ground over the years. Back in 1950s only 5% of the employed people were required to hold a license (Morris M. Kleiner 2010). This five-fold increase in share of licensed workers is not primarily driven by a growing share of workers self-selecting into heavily licensed occupations i.e. changing composition of the workforce. Instead, about two-thirds of the growth of licensed workers comes from increase in the number of occupations that require a license (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Additionally, de-licensing of an occupation rarely occurs, and recent attempts in nine states to collectively de-license groups of occupations have shown more potential but have been almost uniformly unsuccessful (Thornton and Timmons 2015).

The second motivation to study occupational licensing is that it can potentially explain why both geographic mobility and people switching between occupations have been decreasing simultaneously. Since 1980s, economists have documented a secular decline in migration measured at interstate, intercounty and intracounty level, and a decline in labor market transitions - i.e., people are not changing employers (jobs), occupations, and industries as much as they did in the past (Molloy, Smith, and Wozniac 2014). Furthermore, these two stylized US labor market facts cannot be concurrently accounted for by trends popularly studied in the labor literature such as concerns of dual-career households[1], changes in the

1. "Rising share of dual-earner households" which are likely to experience a joint co-location problem with a spouse and are thus likely to stay in a city and not change jobs/industry/occupations only rose from 2% in the 1980s to 3% in the 2000s, which is too small of a change to meaningfully affect aggregate migration patterns.

distribution of job opportunities across space[2], and polarization in the labor market[3] (Molloy, Smith, and Wozniac 2011).

However, if individuals in licensed occupations (who usually make more wages than unlicensed individuals) have a higher opportunity cost of moving out of their licensed jurisdiction, or face additional moving/relicensing costs then occupational licensing is one potential reconcilaitiory factor explaining both why geographical fluidity is decreasing (largely across state boundaries) and why there is less churning (between occupations).

The current large magnitude and high growth of occupational licensing laws in the recent decades has started drawing more policy attention. Occupational licensing has come in the limelight alongside other pressing labor market issues like immigration, changes in demographic structure of workforce, technological advances and globalization of competition. In fact, Institute for Justice concluded in a 2018 report that "occupational licensing is widely recognized as one of the most important labor market issues in the United States, and the topic warrants some more research".

1.2 Some Pros and Cons of Occupational Licensing

Proponents of occupation licensing usually highlight its benefits such as better product and quality control, safeguarding public health and safety through focusing on higher, protecting consumers by guaranteeing minimum educational requirements and industry oversight, supporting career development or pathways for licensed workers, and enhanced professionalism for licensed workers (National Conference of State Legislatures 2019). In particular, they emphasize that occupational licensing can solve the classic lemons problem in service markets and thus avoid market failure due to adverse selection - thereby making labor markets more competitive when forces like reputation or litigation fail to achieve desired outcomes.

However, opponents of occupational licensing often highlight reducing geographic mobility of workers, increasing rent seeking behavior, reducing wages for unlicensed workers rela-

2. "Range of occupations and industries has become more uniform across metropolitan areas and has reduced the incentive of people to migrate.". This only explains the decline in geographic mobility trend but it need not explain decline in labor churning trend.

3. "Expanded use of computers, greater ease of automation and off-shoring increases demand for high-skill jobs (e.g. professional, managerial, and design jobs) but reduces demand for the middle-skill jobs (defined as office administration, sale and production jobs)". Such structural shifts in the occupation distributions can potentially have restrict migration if historically low skilled workers in food service, personal care services or cleaning services are more likely to move to a different labor market to take middle-skill jobs, and there was an elimination of large shares of middle-skill jobs that could limit the "migration worthy" jobs for less educated workers. Unfortunately, again there is no not much empirical support for this "hallowing-out effect" hypothesis.

tive to their licensed counterparts, reducing employment in licensed occupations, reducing market competition and innovation, increasing the price of goods and services, and disproportionately burdening immigrants, military veterans and their families, people with a criminal history[4] and dislocated or unemployed workers (National Conference of State Legislatures 2019In particular, they have argued that occupational licensing goes against free market principles as it restricts movement of labor across the country, depriving workers of opportunities not only in their current state of residence but also other "greener economic pastures" (Mulholland 2016).

Since occupational licensing may hurt the economy of gains from trade in labor services that are based on comparative advantages or variation in market opportunities, some today are raising the call for occupational licensing system reforms (Mulholland 2016; Sanders 2020).However, it is interesting to note that de-licensing of an occupation rarely happens, and attempts in nine states to collectively de-license groups of occupations in 2015 has been almost uniformly unsuccessful (Thornton and Timmons 2015).

1.3 Certification versus License

The three terms "license", "certification", and "registration" are often used interchangeably causing confusion, but have more formal and very specific meanings.

Registration usually requires individuals only to list contact information with a specific agency, usually a designated government body, and as a general rule does not require the person to pass an exam or meet hold pre-determined standards (Sanders 2020).Thus, they are the least restrictive regulation out of the three basic forms. Certification on the other hand give the holder a protected occupational title, and individuals without the certificate may practice the occupation as long as they do not use the protected title. However, a license restricts individuals from indulging in "scope of practice" without permission from a government agency, and can be used to refuse people the legal opportunity to earn a living in a chosen field (Sanders 2020).

More formally, according to Current Population Survey (CPS), certifications and licenses are both non-degree, time-limited credentials that need to be renewed periodically, and demonstrate a level of skill or knowledge needed to perform a specific type of job (Labor Statistics 2017).

The difference between the two is based on who issued the credential. More specifically, "certification is a credential awarded by a non-governmental certification body based on an

4. 1 in every 3 American adults has a criminal record, as per Bureau of Justice Statistics, 2016.

individual demonstrating through an examination process that they have acquired the designated knowledge, skills, and abilities to perform a specific job" (Labor Statistics 2017). In particular, certificate does not convey a legal authority to work in an occupation. For example, Project Management professional certifications by Northeastern University, or Python/R data science certifications on LinkedIn. A certification can be highly valuable too, like Charted Financial Analyst (CFA) designation.

In contrast, "license is a credential awarded by a governmental licensing agency based on pre-determined criteria, which could include some combination of degree attainment, certifications, educational certificates, assessments (including state-administered exams), apprenticeship programs, or work experience" (Labor Statistics 2017). More importantly, license conveys a legal authority to work in an occupation. For example, a K-12 teaching license, a nursing license or even a cosmetology license.

Current Population Survey started collecting data on these two non-degree credentials from 2015, given its paucity and their distinct nature from traditional educational attainment data, like a high school degree or a bachelor's degree, to help policymakers/ researchers study how these two types of credentials affect labor market success.

1.4 Paper Overview and Organization

In this paper, I use three recently added non-degree credential questions in Current Population Survey to estimate whether a person is licensed by the state government to work in their primary occupation or not from 2016 to 2020. I have more than 400 unique occupations spread across 50 US states and Washington, D.C.. Within an occupation-state at a given period of time, all individuals should either be licensed or unlicensed. However, I find that the individual license estimate is noisy, and thus use k-means unsupervised machine learning algorithm to correct for the misreporting in survey, and estimate an individual's licensing status in both current and last year. The algorithm suggests that within a state-occupation-time cluster where less than 22% individuals report they are licensed, everyone should be unlicensed. Similarly, in occupation-state-time clusters where more than 69% individuals report they are licensed, everyone should be licensed.

I find individuals in k-means suggested licensed occupations do have higher wages, just like theory and past empirical work suggests, and this serves as a sanity check. Then, I run a state-occupation-time fixed effects model with detailed individuals demographic controls and find that occupation licensing does not have a negative impact on worker geographic mobility between states, and occupational licensing also does not impact geographic mobility within the state. This preferred specification is quite robust to choice of controls and alternative specifications.

The rest of the paper is organized as follows:

Section 2 discusses literature, in particular the four outcome variables that have been typically examined in occupational licensing literature and specific geographic mobility results.

Section 3 is about the data and describes the two data sources 3.1, provides an overview of data transformations required to be able to run the preferred specification 3.2, details the three certification questions and construction of the licensing variable from them 3.3, corrects the misreporting in the self-reported licensing variable using k-means unsupervised machine learning algorithm 3.4.1, changes the interpretation of migration outcome variable from an in-migration specification to my preferred out-migration specification 3.4.2, and provides the summary statistics 3.5.

Section 4 describes the identifying equation and the fixed effects model preferred specification. Section 5 presents and discusses the results, while Section 6 concludes.

2 Literature Review

2.0.1 Historical Occupational Licenisng Study Categories

Occupational licensing studies in the past have mostly looked at individual occupations and studied them in depth in one the following four distinct classes of outcome variables, namely - (1) Impact on Quality, Health and Safety, (2) Impact on Wages and Employment, (3) Impact on Consumer Prices, and (4) Impact on Geographic Mobility (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015).

First, occupational licensing studies on quality, health and safety are usually motivated by how occupational licensing can potentially reduce consumer uncertainty about the labor service quality, and thereby increase consumer's demand for the licensed service (Arrow 1971). For example, if it takes consumers too much time, effort or they simply cannot cannot distinguish between high and low quality electrical work, then competition may crowd out the high quality electricians from the market and only the "lemons" or low quality electricians will serve the entire market. Thus, while occupational licensing can theoretically fix the lemon problem, most empirical evidence does not find that stricter licensing requirements improve these outcomes. For example, teaching licenses usually have no effect on student scores

(Kane 2008),tighter dentistry licenses requirements have no effect on dental deterioration or amount of repair required (Kudrle 2000), and floristry license requirement has no effect on floral arrangements ratings by florist judges (Carpenter 2012).

Second, occupational licensing studies on wages and employment usually motivated by regulatory capture theory, where practitioners capture monopoly rents, restrict labor supply and drive up prices of labor. Thus, a wage gap gets created between those who manage to enter licensed occupations and those who fail to enter the licensed occupations, usually estimated to be about 5 to 10 percent (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Also, the wage premia differs by occupations with different licensed occupations experiencing different levels of wage growth, and the highest wage growth is documented in some of the occupations that also have the greatest proportions of licensed workers, such as health care support and practice, education, and legal occupations (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Furthermore, employment grows at a slower rate in licensed occupations. For example, states with the least restrictive regulations of nurse practitioners (NPs) had more licensed NPs and patients were more likely to receive primary care from NPs when physician primary care was lacking (Goodwin et al. 2013).

Third, occupational licensing studies on consumer prices theorize that prices should be higher as supply of new professionals is reduced and competition between providers is less vigorous. Many studies find that more restrictive licensing laws lead to higher prices for consumers. For example, regulating the number or functions of mid-level dental-care providers (hygenists and assistants) increases price of filling a cavity or dental visit (price index of 12 services) (Liang and Ogur 1987; Isom 2019) , additional state mortgage broker bonding/net worth increases the probability that mortgage is high priced (Kleiner 2007), and medium or high level of nursing license regulation increases the price of well-child medical exams (Marier et al. 2014).

Fourth, and finally, occupational licensing studies on geographic mobility are usually motivated by costs associated with re-allocation which may prevent migration. For example, additional monetary and non-monetary time costs like time to meet new qualifications in destination state, such as education, experience, training or testing, can discourage individuals from relocating. Alternatively, if the opportunity costs are higher for licensed individuals to move out of their home state compared to non-licensed individuals say because licensed individuals have higher wages, we can also see lower geographic mobility for people in licensed occupations as a result.

2.1 Some Occupational Licensing Studies on Geographic Mobility

First, there has been relatively little research on effects of occupational licensing on geographic mobility (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015), and the little research is largely descriptive in nature because of the lack of available information on changes in licensing requirements that could be used in a traditional causal framework (Johnson and Kleiner 2020). Also, there was a shortage of data on non-degree credentials till about 2010, and Kleiner and Krueger (2010) conducted the first national study using specially designed Gallup survey to study occupational licensing (though their study focused on wage dispersion).

Second, in the past, emphasis has been on a few occupations like nurses and lawyers only, while many other occupations have been largely not studied. For example, lower interstate mobility of lawyers could be traced to investments in state specific law and procedure (estimated to have a small effect), and to licensing through state bar exam (Pashigan 1977).

More recently, as states started adopting reciprocity agreements (which lower re-licensure costs), occupational licensing‘s impact on lawyers and nurses has been studied through a difference-in-difference design. However, we get mixed results across occupations. For instance, mobility of nurses did not increase following staggered adoption of Nurse Licensure Compact[5] (NLC) by states, even among residents of counties bordering NLC states that should be most impacted (DePasquale and Stange 2016). On the other hand, there is some evidence that the adoption of reciprocity agreements increases the interstate migration rate of lawyers (Janna E. Johnson 2017).

Furthermore, while universal licensure[6] is gaining some popularity now, with as much as 11 states passing laws to mandate their occupational boards to allow individuals with out-of-state licenses to obtain a valid occupational license to practice in 2021, there are still significant limitations (Follett, Hentze, and Herman 2021). Almost always this process is not automatic, and not only do licensees must meet certain residency, testing, background check and other requirements, but as well as pay all applicable fees in order to practice in the new state.

Third, apart from certain occupations, geographic mobility occupational licensing studies examine certain groups of people like military spouses and immigrants. Military spouses

5. An interstate compact that allows registered nurses and licensed practical and vocational nurses to practice across state lines in participating states without acquiring a new license.

6. Defined as a state recognizing a person’s occupational license granted in another state as valid.

often struggle to find and maintain employment due to frequent moves (an average of once every three years), and are particularly vulnerable to licensing requirements (Hentze 2021). Similarly immigrants, described as "grease on the wheels of the labor market" since they help labor market efficiency by making geographic choices based on wage differences across locations, are adversely affected too (Borjas 2001). Licensing reduces the mobility of immigrants, may exacerbate worker shortages and can slow the entire economy (Cassidy 2019). Even low-skilled immigrants workers are affected and disperse less as a result of occupational licensing regulations, as evidence from from Vietnamese immigrants who came to US after Vietnam War and entered the manicure industry (Federman 2006).

Fourth, there have been very few attempts to generalize occupational licensing geographic mobility results to all occupations. In 1977, Pashigan found that occupational licensing has had a quantitatively large effect in reducing interstate mobility of 34 professionals occupations (Pashigan 1977). More recently, there are only three works that I am aware of. Out of these, Kliener and Johnson (2020) do the most rigorous study and single out causal effects of occupational licensing on interstate migration by disentangling the decision to move into three components - risk aversion, loss of local capital, and relicensing costs.

To get to the casual effects, Kliener and Johnson (2020) assume that moving more than 50 miles destroys all local clientele/business reputation effects, and people who stay in their home state are fundamentally higher risk averse individuals who value family ties and friendships much more (Johnson and Kleiner 2020). Thus, by throwing out these two sub-groups of people, they can look at actual costs of occupational licensing on interstate migration as occupational licensing in US is primarily at state level. Furthermore, Kliener and Johnson (2020) prefer to rely on American Community Survey over Current Population Survey (CPS) due to three reasons, namely - (1) CPS does not have sub-state place of residence (required to disentangle local client effect), (2) CPS does not report state of birth, but only country of birth (required to disentangle risk aversion), and (3) CPS has much smaller sample size (Johnson and Kleiner 2020)[7].

Therefore, using American Community Survey data from 2005 to 2018 and looking at individuals who move 50 or more miles and reside outside their state of birth, their preferred specification shows that individuals in 16 state-specific universally licensed occupations[8] move between states at a 7 percent lower rate compared to members of 6 quasi-national

7. One advantage of Current Population Survey over American Community Survey is that it allows one to differentiate between new entrants and continuing members in an occupation.

8. Example - bar for lawyers or the pharmacy jurisprudence exams, which are state-specific but nonetheless these occupations are licensed in all states.

licensed occupations[9] (Johnson and Kleiner 2020).

They get similar results when they use Current Population Survey as a robustness check and compare interstate migration in the 16 state-specific universally licensed occupations (treatment group) against all other occupations (excluding 6 quasi-licensed occupations). This specification is close to my empirical work. However, to estimate who is licensed and who is not, they simply assume that only individuals in the 16 state-specific universally licensed occupations are licensed from 2005 to 2018. Thus, assuming licensing is not increasing over time is a very strong assumption, especially given that we know that new occupations tend to get licensed. I improve upon this licensing variation by using the three new certification questions added questions in Current Population Survey from 2016 to 2020 and k-means algorithm to better capture the growth in occupational licensing over time and states. Current Population Survey comes with certain disadvantages too though, namely that disentangling different components of migration cannot be determined.

Second, a White House report in 2015 that tried to replicate upcoming Kliener and Johnson (2020) approach found adverse (state) occupational licensing restricting interstate migration too. Using American Community Survey data from 2010 to 2013 and looking at people from age 25 to 65, they found that interstate migration rates for workers in the most licensed occupations are lower by almost 15% of the average migration rate compared to those in the least licensed occupations Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Difference in within-state migration rates for workers was much smaller - only about 3% of the average migration rate. These impacts were also much larger for younger (under 35) licensed workers. Furthermore, the using in Survey of Income and Program Participation (SIPP) over the eight-month period starting in late 2012, licensed workers were about 20% less likely non-licensed to change states Treasury Office of Economic Policy, Economic Advisers, and Labor 2015).

Third, Nunn (2016) used the three certification questions in Current Population Survey to look at how occupational licensing affects wages, employment and migration (within state and across state moves of individuals). He presents simple graphs displaying his results but does not report statistical significance or regression tables. After adjusting for observable individual[10] characteristics (work experience, education, gender, and race), licensed workers appear somewhat more likely than their unlicensed counterparts to move within a state, but are much less likely to move outside their state (Nunn 2016). In particular, he does

9. Occupations that have a national exams with a single passing standard, such as for nurses, physicians, and social workers who must also transfer their license when they move between states, although at a slightly lower cost.

10. Aged 25 to 64

not use geographic/state or occupation fixed effects, which are likely driving his results[11]. Furthermore, Nunn (2018) published using the new data in Current Population Survey again to look at occupational licensing, this time with a richer set of individual level and geographic controls, but only focuses largely on the wage margin, operationalised by wage distribution, premia, inequality, and omits the geographic mobility/migration outcome this time (Nunn 2018).

Thus, there is still some value in using the new Current Population Survey data to study occupational licensing and its impact on worker geographic mobility.

3 Data

3.1 Sources

I obtain data from two surveys - Current Population Survey (CPS) and the Annual Social and Economic Supplement (ASEC).

CPS is a representative sample of the civilian household population in the US surveying occupants of about 60,000 households every month, and follows a 4-8-4 rotation pattern where every household/address is interviewed once a month for 4 months straight and then interviewed again after a gap of 8 months (1 year later) in order to reduce the inconvenience to each household while allowing for some yearly or monthly comparisons (U.S. Bureau of Labor Statistics 2019). This in-person and telephonic interview was launched in 1940s and continues till today, providing many official labor force statistics for the US as it collects both labor market health and demographic data, including on income and poverty issues.

While CPS collects labor force data on a monthly basis, additional topical data are often collected via supplements to the monthly survey. Annual Social and Economic Supplement is one premiere supplemental survey to Current Population Survey which is annual (as the name suggests) and the ASEC data is mostly collected in March. Thus, ASEC is also called the March Supplemental to the Basic Monthly March CPS, and is in fact a widely used dataset.

11. My own work shows the importance of controlling for these, and occupation fixed effect is even more important than geographic/state fixed effect term. Since he published as soon as the first round of certification data was released, he does not need time fixed effect

3.2 Overview

The data section has three important ideas - construction of occupational licensing variable from three certification variables and correcting this noisy measure with k-means clustering algorithm, merging of CPS and ASEC records, and finally transforming migration variable interpretation in our clean data to form an out-migration specification instead of an in-migration specification (a slightly more interesting hypothesis). In this subsection, I will provide a brief overview of why these three steps are required, and then talk about each transformation in more depth later in the separate subsections.

First, I merge individual records across the two datasets (for March of every year) because our outcome variable (migration status) exists only in ASEC, while our key variable of interest (self-reported occupational licensing status) can only be constructed from three certification questions added in CPS.

Also, ASEC sample size is larger than CPS by construction, and thus I use Current Population Survey as the master data[12]. I merge March Basic Monthly CPS survey (master data) with March Supplemental (Annual Social and Economic Supplement) using "MARBASEID"[13] identifier constructed by IPUMS[14] to generate our pooled cross section data (Flood and Pacas 2018.) Only less than 5% people in CPS are not matched with people in ASEC, and 281,428 individuals are matched. I drop occupations that have less than 10 individuals, and am left with 281,404 individuals. Section 3.4 will discuss the Current Population Survey master data cleaning process, along with the merge results with ASEC in more depth.

Second, using the three certification questions in CPS, I create the key parameter - active government licensing, which informs us if an individual is licensed by the state government or not to practice their primary job in their occupation within the state. Construction of this variable is discussed in Section 3.3.

12. CPS has been expanded over the years to get better estimates on certain sub-populations, like persons of Spanish (Hispanic) origin and low-income children who do not have health insurance coverage to increase reliability of estimates and help answer certain policy question. However, ASEC sample had to be expanded at a faster rate to supplement the CPS, in particular due to people dropping out of CPS in lieu of its inherent 4-8-4 rotation pattern.

13. "MARBASEID" was created specifically to correctly link the ASEC and the March Basic Monthly data and saving researchers duplication effort. The match is otherwise complex for two reasons (Flood and Pacas 2018). First, the variables required to link the ASEC to CPS monthly files are not available for all years on the ASEC. As a result of omitted linking keys and the ASEC oversample, duplicate and false matches are problematic. Second, Census Bureau transition to a computer-based interview resulted in more prominent data quality issues for linking across months even if they did not compromise the integrity of each individual month of data.

14. A leader in the production of easily accessible, population-level data. www.ipums.org

Furthermore, I correct for misreporting within occupation-state clusters, as everyone within an occupation-state cluster at a given period of time should ideally give a uniform answer, either yes or no, about being licensed by the state. Since I find that even within occupations that we know are licensed by all states, like registered nurses where only 82% people report they are licensed, and this misreporting occurs to varying degrees in all states, I conclude that the active government license variable generated from the three certification questions is a noisy measure of being licensed (by the state in a particular occupation). Thus, I reduce the misreporting error by using k-means clustering algorithm to determine which occupations-state clusters are licensed or not at any given period of time.

In particular, I collapse individual level license dummy variable (281,404 observations) in the merged CPS ASEC data by state, occupation and time to get proportion of people who are licensed in each occupation-state-time clusters (58,117 observations)[15]. Then, I classify each of these 58,117 collapsed state-occupation clusters into 3 categories (licensed, unlicensed and not sure) or 2 categories (licensed, unlicensed).

When I created 3 categories, k-means algorithm suggested state-occupation-time clusters where less than 22% people have an active government license are unlicensed occupation-state-time clusters, and state-occupation-time clusters where more than 70% people have an active government license are licensed state-occupation-time clusters. The in-between state-occupation-time clusters are put in the 'not sure' about state-occupation license status category. Thus, I standardize or correct for the individual license status (active government license variable response) within all state-occupation-time clusters, or for all 281,404 individuals in the dataset.

Similarly, if I create only 2 categories, k-means algorithm suggests that state-time-occupation clusters where less than 42% people have an active government license should be non-licensed state-time-occupation clusters. Regardless of creating 2 or 3 cluster classifications, the k-means algorithm correction reduces the number of people who have an active government license slightly. More details will be discussed in Section 3.4.1.

Third, I prefer to test that if one is currently licensed by the state, they did not move to another location over the next one year (out-migration specification), rather than to test if one is currently licensed by the state, they moved from another location over the last one year (in-migration specification).

15. Note that I have 5 years of data (2016 to 2020) with 51 states and 442 unique occupations, so the maximum number of occupation-state-time clusters that could have been formed is 112,710. However, the actual data only has 58,117 occupation-state-time clusters because there are occupations that have few people and thus may not span across all states, and at time not even across all years.

The default migration question in ASEC asks individuals if they moved from abroad, moved from another state, moved from another county within state, moved within a county, or did not move over the last one year, which without any further transformations only allows me to test the in-migration specification. Moreover, since ASEC provides not only individual's current occupation and state of residence but also individual's last year occupation and state of residence too, I can test the out-migration specification only if I can estimate the individual's last year licensing status too[16]. Here, k-means algorithm allows me to estimate individual's last year licensing status too, and thus get to my preferred specification.

In particular, I apply k-means algorithm decision rule to identify which individuals were licensed last year only when I have licensing data on these individual's last year occupation-state-time cluster in my data. Since the certification questions data in CPS was only released from 2016 onward, I cannot estimate last year licensing status for individuals in 2016, and lose one full year of data. Similarly, I lose a few more people because there was no one in their last year occupation-state-time cluster who had answered the three certification questions. Section 3.4.2 will discuss this, and I will finally be left with about 200,000 individuals when I create licensed/unlicensed classification, and about 157,000 individuals when I create licensed/unlicensed/do not know classification.

The rest of the section is organized as the following. I will first elaborate about the construction of our occupational licensing variable in Section 3.3, then about Current Population Survey data cleaning process and merge results with Annual Social and Economic Supplemental in Section 3.4, followed by k-means correction in Section 3.4.1, transforming the dataset to generate the preferred migration interpretation in Section 3.4.2, and finally present the summary statistics in Section 3.5.

3.3 Construction of Occupational Licensing Variable

CPS added 3 certification questions in its monthly surveys since January 2015 based on the work of the Interagency Working Group on Expanded Measures of Enrollment and Attainment (GEMEnA[17]) (Allard 2016).

Together, these 3 questions (profcert, statecert, jobcert) allow us to create a proxy for whether a person requires license to operate in a particular occupation within a state or not. The questions were -

16. The three certification questions in CPS are monthly
17. Members included senior staff from the U.S. Bureau of Labor Statistics (BLS), the Census Bureau, the Council of Economic Advisors, the U.S. Department of Education's Office of the Under Secretary, the National Center for Science and Engineering Statistics, and the OMB.

3.3.1 Profcert

1. *"Do you have a currently active professional certification or a state or industry license? Do not include business licenses, such as a liquor license or vending license."*

The first question (profcert) is asked of everyone above 15 in Current Population Survey, excluding those with an employment status of "Armed Forces" (as they are not part of civilian institutional population).

Second, profcert is used to identify people with certifications or licenses (working definitions for certifications and licenses are in Section 1, not business licenses issued to companies. Examples of occupational licenses issued to individuals giving them the right to practice include a real estate license, a medical assistant certification, a teacher's license, or an IT certification (Thornton and Timmons 2015).

In particular, some credentials that are not considered a certification or license in the Current Population Survey include - (1) Licenses to engage in personal, as opposed to professional, activities. For example, regular driver's licenses are excluded, but commercial driver's licenses (CDLs) are included. (2) Licenses issued to businesses, such as liquor licenses or vending licenses. (3) Certificates of attendance or participation in short-term training. (4) Educational certificates awarded by an educational institution—such as a community college, a 4-year college or university, or a trade school—based on completion of all requirements for a program of study. While this training may help in the performance of a specific job, educational certificates are not necessarily required or considered proof of qualification. These types of credentials are not time-limited. Examples of educational certificates include a digital arts certificate from an online university or a motorcycle mechanics diploma from a community college (Labor Force Statistics from the Current Population Survey, April 2017).

Third, people may have more than one of these credentials.

Lastly, note that profcert does not distinguish between certifications or licenses.

3.3.2 Statecert

2. *"Were any of your certifications or licenses issued by the federal, state, or local government?"*

The second question (statecert) is asked of people who answered "yes" to the first question. According to the definition developed by The Interagency Working Group on Expanded Measures of Enrollment and Attainment (GEMEnA), licenses are issued by a governmental body and people who answered "yes" to the second question are classified as having a license.

People with a license may also have a certification.

Four things to note about statecert.

First, the intent of statecert question is to determine the issuing authority of a credential and thus, to distinguish between certifications and licenses. Recall licenses are issued by a governmental body, unlike certificates. It is natural to ask this question after profcert because of the interest in distinguishing between certifications and licenses.

Second, while statecert does not help us distinguish between the level of government that issued the license, we know that most occupation licensure occurs at state level and not local or federal level. To substantiate this claim, I will quote Federal Trade Commission (2018) - "Occupational licensing, which is almost always state-based, inherently restricts entry into a profession and limits the number of workers available to provide certain services. It may also foreclose employment opportunities for otherwise qualified workers. This reduction in the labor supply can restrain competition, potentially resulting in higher prices, reduced quality, and less convenience for consumers..." (Goldman 2018).

Third, statecert is a conditional question. Only those who answered yes to profcert were asked this question. In other words, if a person answered no to profcert, they are not asked statecert question.

Fourth, statecert was modified to allow for the situation in which a person has more than one certification or license. Instead of "was it issued by ..." the question wording was modified to "were any of your certifications or licenses issued by ..." to allow for the possibility of multiple credentials. Such modification rules ensure questions are not misunderstood, and complex skip patterns using the responses to several questions answered in the past ensure that CPS interview stays short and clear.

3.3.3 Jobcert

3. *"Earlier you told me you had a currently active professional certification or license. (Is/was) your certification or license required for your (job/main job/job from which you are on layoff/job at which you last worked)?"*

The third question (jobcert) is asked later[18] in the Current Population Survey interview

18. After consideration, Bureau of Labor Statistics (BLS) decided that placing the first two questions after the educational attainment questions made the most sense. The primary reason behind this decision was the importance of collecting information from these two questions for all individuals, not just for those who were employed. Unlike the job-related questions, the educational attainment questions are asked of everyone. The third question (jobcert), which asked specifically about whether the credential was for the respondent's job, did belong among other job-related questions.

of employed and unemployed people who were identified in the first question (profcert) as having a certification or license.

First, jobcert is also a conditional question like statecert but subject to more constraints - it has the smallest universe of out of the three certification question. In particular, unlike profcert and statecert, people with employment status of “Unemployed new workers” and “Not in the Labor Force” (NILF) (due to “Retirement”, “Inability to work”, or “Other reasons”) are not asked jobcert question. Thus, jobcert is only asked from everyone who answered “yes” to profcert question and falls in one of the three employment categories - “At work”, “Has job, not at work last week”, or “Unemployed experienced worker”. Current Population Survey subject matter experts decided that it was important to ask jobcert of the unemployed about their past jobs. This is consistent with other labor force questions, such as industry and occupation, and allows for an examination of the unemployed who have previously worked.

Second, in jobcert question, “required” can mean either legally required (such as a license to practice medicine) or required by their employer (such as a computer maintenance certification). Some people may have credentials that they do not use for their current job, such as a computer technician who obtained a real estate license for personal interest or for a second job. Also, people sometimes have certifications or licenses that may be helpful for their jobs but are not required by law or by their employer. Thus, jobcert question allows us to create a strong working level, binding definition for our licensing variable.

Third, jobcert question wording differs by individuals’ labor force status/employment. Thus, the question is not as complex as it may initially seem. For example, people who are employed at only one job are asked about their job, while those with multiple jobs are asked about their main job. Likewise, unemployed individuals who are on layoff are asked about the job from which they are on layoff, and unemployed individuals who previously worked are asked about their last job.

Finally, we can construct an individual self-reported licensing status measure, active government license if a person answered yes to statecert and yes to jobcert. By construction, this is equivalent to if a person answered yes to profcert, statecert and jobcert. Table 1 above reports the results.

Since Current Population Survey is a high-quality national level survey implementing many

Table 1: Summary Statistics of Certification Questions

	Mean	Std. Dev.	Min	Max	Universe
Profcert: Has active prof-cert/license	0.251	0.43	0	1	281404
Statecert: Has govt issued prof-cert/license	0.907	0.29	0	1	70584
Jobcert: Cert required for job	0.839	0.37	0	1	70584
Licensed / Govt. issued prof-cert required for job	0.196	0.40	0	1	281404
Observations	281404				

Source: Current Population Survey (CPS), 2016-2020

Profcert Universe: 16-79 years old excluding people employed in Armed Forces.

Statecert Universe: Everyone who answered “yes” to Profcert. Those who answered “no” to Profcert are excluded.

Jobcert Universe: Everyone who answered “yes” to Profcert, excluding people “Not In the Labor Force” (NILF) and “Unemployed New Workers”. Again, those who answered “no” to Profcert are excluded.

best practices/wording choice when collecting data, I do not worry about serious misreporting. In fact, testing in about 25 cognitive interviews where respondents included individuals who had certifications or licenses as well as those who did not revealed no problems with the proposed placement or wording of the questions. Also, the cognitive testing did not find evidence that the new questions had an impact on responses to existing questions. Thus, Bureau of Labor Statistics decided to accept the proposal to permanently add the three questions about certifications and licenses to the monthly Current Population Survey. Furthermore, non-response on the questions is quite low, and regular imputation method used for other Current Population Survey questions was used on the three certification questions.

Nonetheless, examining the raw data/individual responses suggests there is in fact some misreporting. Note again that occupational licensing happens mostly at state level, so our key variable active government license can be interpreted as State Occupational Licensing. Thus, all individuals within a state-occupation-time should either answer “yes” or “no” unanimously, but instead we find that even in state-occupations that we know require a license for individuals to operate, not everyone answers correctly that they have a license. For example, even in occupations that that are licensed in all states like “Registered Nurses”, only about 82% registered nurses at the national level report they have an active government issued license (see Table 2 below), and the misreporting is spread across states (see Figure 1 below).

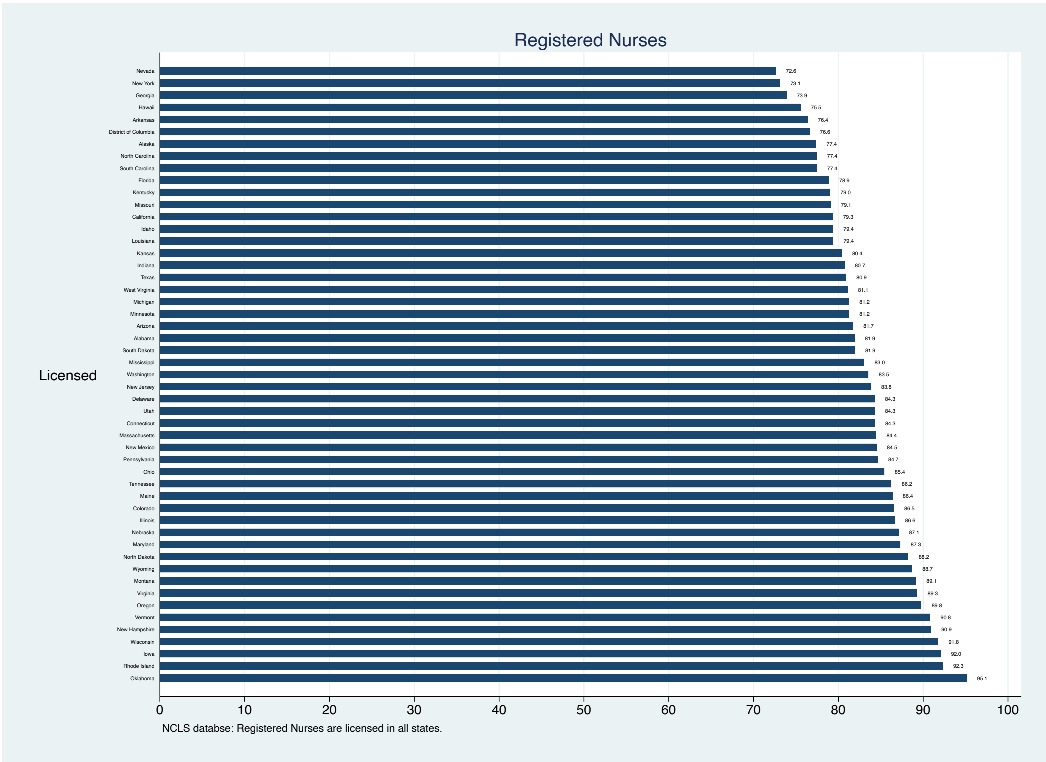
Thus, there may be some under-reporting on average for occupations that are licensed and I

Table 2: Proportion of People Licensed in Occupations (ranked from top 20 to lowest 20 occupations)

Occupation	Proportion Licensed	Frequency
Chiropractors	.915	118
Optometrists	.899	79
Dentists	.879	282
Veterinarians	.876	170
Physicians and Surgeons	.857	1853
Pharmacists	.847	600
Occupational Therapists	.845	238
Registered Nurses	.825	6212
Lawyers, and judges, magistrates, and other judicial workers	.805	2411
Physical Therapists	.804	484
Aircraft Pilots and Flight Engineers	.802	248
Respiratory Therapists	.794	194
Physician Assistants	.791	196
Speech Language Pathologists	.786	294
Emergency Medical Technicians and Paramedics	.78	414
Psychologists	.774	389
Dental Hygienists	.754	357
Special Education Teachers	.741	729
Radiation Therapists	.735	34
Secondary School Teachers	.731	2093
...	...	...
...	...	...
...	...	...
Bank Tellers	.0071	567
Postal Service Clerks	.0049	203
Laundry and Dry-Cleaning Workers	.0033	303
Reinforcing Iron and Rebar Workers	0	10
Tire Builders	0	11
Communications Equipment Operators, All Other	0	14
Avionics Technicians	0	14
Paving, Surfacing, and Tamping Equipment Operators	0	20
Food & Tobacco Roasting, Baking, and Drying Machine Operators & Tenders	0	21
Textile Knitting and Weaving Machine Setters, Operators, and Tenders	0	21
Gaming Cage Workers	0	21
Rolling Machine Setters, Operators, and Tenders, metal and Plastic	0	21
Adhesive Bonding Machine Operators and Tenders	0	21
Helpers–Installation, Maintenance, and Repair Workers	0	26
Prepress Technicians and Workers	0	38
Textile, Apparel, and Furnishings workers, nec	0	40
Extruding & Drawing Machine Setters, Operators, and Tenders, Metal & Plastic	0	41
Credit Analysts	0	48
Pressers, Textile, Garment, and Related Materials	0	54
Extruding, Forming, Pressing, & Compacting Machine Setters, Operators, & Tenders	0	54
Ushers, Lobby Attendants, and Ticket Takers	0	76

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Figure 1: Proportion of Registered Nurses reporting holding a License, by state. Registered Nurses are licensed in all states, yet individuals working in this occupation report they do not have a license.



will correct for this using the following methodology - if a high proportion of people within the state-occupation-time cluster hold an active government license, we can infer that the occupation is licensed by the state, and assign everyone in the state-occupation-time cluster with an active government license value of 1. Similarly, if a low proportion of people within the state-occupation-time hold an active government license, we can infer that everyone in the state-occupation-time does not hold a license. I discuss about this correction methodology in Section 3.4.1 below.

3.4 Data Cleaning and Transformations

First, I start with Current Population Survey survey as my master dataset, and apply three filters (year, employment status and age) so that we can start with a sample of civilian non-institutionalized working-age population which could be asked all three certification questions.

Firstly, I keep year 2016 onwards only (even though the three certification questions were introduced in January 2015) because jobcert question's responses are not reported for 2015[19], and exclude Current Population Survey data for 2021 as 2021 Annual Social and Economic Supplement data is still not released by IPUMS.

Secondly, people with employment status of "Armed Forces" are dropped since they are asked none of the three certification questions (part of institutionalized population). Furthermore, people with employment status of "Not In Labor Force (NILF)" and "Unemployed New Workers" are dropped too as they are not asked jobcert question (but are asked the other two). It does not make sense to ask them about jobcert as they do not have a job yet (or much job history) - they are marginally attached with the labor market.

Thirdly, I restrict our analysis to a working age population, defined as ages 16 to 79. Current Population Survey asks these three questions about certifications and licenses of all people in the survey age 15 and older, but individuals between the ages of 80-84 are clubbed as 80 while anyone over 85 is clubbed as 85. (Results are robust to working age population definition of 16-65, and 25-55).

Thus, everyone in our sample is in the labor force and a potential candidate to answer all three questions asked the question (though everyone may not be asked statecert or jobcert due to the conditional nature of questions - respondents have to answer "yes" to profcert).

Second, I then create the primary (dummy) variable of interest named active government license which identifies if a person has an active government issued license to practice the profession, as discussed in Section 3.3 above in the Current Population Survey dataset. Again, our definition is if a person answered "yes' to statecert and "yes" to jobcert. By construction, these people also answered "yes" to profcert question.

Third, I merge Current Population Survey data (master) with ASEC data using the individual unique id created by IPUMS called "MARBASECID". Only about 12,681 observations remain unmatched in the master dataset, and 281,428 individuals are matched. Thus, only about 4% of data did not match between Current Population Survey (March Basic) and Annual Social and Economic Survey (March Supllemental) for every year in the time period 2016-2020[20].

Most individual level controls come from Current Population Survey data such as age, gender, race, marital status, number of own children under 5 in household, but ASEC data contains

19. In 2015, the first year of data collection, jobcert question was not asked in May and June 2015 of households in their first and fifth interviews due to a programming error, leading to lack of data for these 2 months

20. It may be possible to increase the match count even further by trying to marge March Supplemental data with Current Population Survey February Basic and Current Population Survey April Basic data.

year on year migration status (used to construct outcome variables), and total personal income data. Thus, I had to merge the two datasets.

Fourth, I drop occupation categories that have less than 10 people as they may be more prone to misreporting error due to the occupation group being formed out of less number of people. 24 observations falling in 4 occupations get dropped.

Fifth, I will now create a new measure of active government license that corrects for misreporting in license status within an state-occupation-time cluster so that all individuals within an occupation-state cluster at a period of time are either licensed or not (i.e. have uniform replies). Subsection 3.4.1 below talks about the process in depth, and I will end this section by talking about estimating the active government license status last year for individuals to change the migration variable interpretation in Section 3.4.2. Finally, I show summary statistics in Section 3.5.

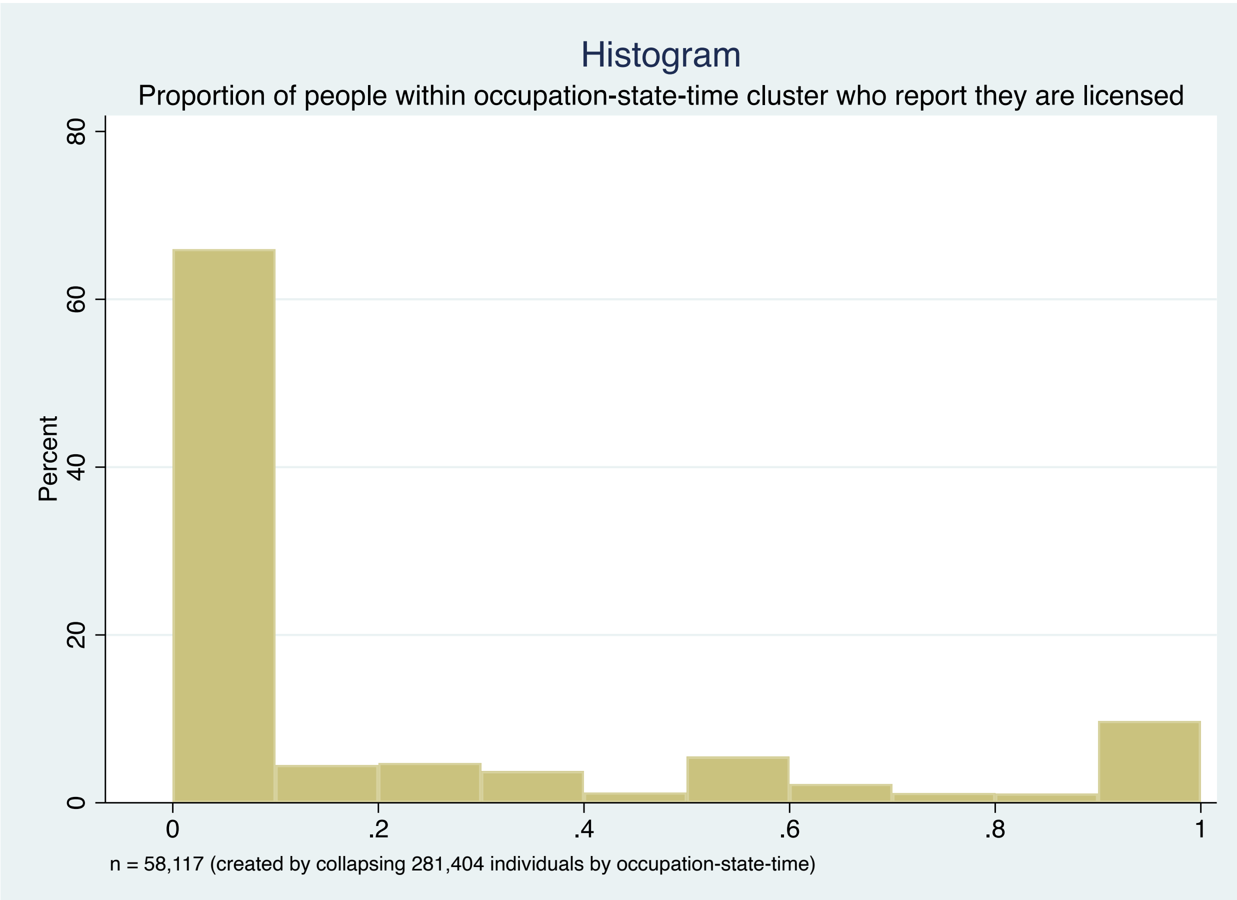
3.4.1 Correcting Mismeasurement in Active Government License variable using k-means algorithm

As mentioned in the Section 1, licensing has been increasing over time, though in our raw data time period from 2016 to 2020, licensing has been increasing very slowly. Moreover, de-licensing of an occupation rarely occurs (Thornton and Timmons 2015). Nonetheless, to be safe I control for the time dimension as some states could be switching occupation from unlicensed to licensed status over this time frame. Thus, I temporarily collapse the merged individual level data to state-occupation-time level (281,404 individuals collapsed into 58,117 state-occupation-time clusters) to examine the distribution of active government license dummy at state-occupation-time level (Figure 2).

The interpretation of the collapsed license dummy at state-occupation-time level is the following: proportion of people within an occupation-state-time that are licensed, and I use the distribution of these proportions using k-means algorithm to generate a decision making process so that I can unbiasedly conclude which occupations-state-time clusters are likely licensed or not, and thus rectify individual misreporting by superimposing the group wisdom on their individual "wrong" answers.

The decision making process is based on the simple idea - if a low proportion of people within a occupation-state-time cluster report they are not licensed, it is likely that nobody is licensed within that occupation-state-time cluster. However, if a high proportion of people within a occupation-state-time cluster report they are licensed, it is very likely that everyone within that occupation-state-time cluster is licensed.

Figure 2: The maximum number of occupation-state-time clusters that we can have is 112,710 - since our data has have 5 years (March month only from 2016 to 2020), 51 unique states and 442 unique occupations. Since some occupations may have relatively few number of people not spread across all states (or even years), the actual number of occupation-state-time clusters in our sample is smaller.



To chose the cutoff values for proportion of people within the occupation-state-time cluster beyond which everyone should be imputed a value of 1, and under which everyone should be imputed a value of 0, I use a simple unsupervised machine learning clustering algorithm called k-means.

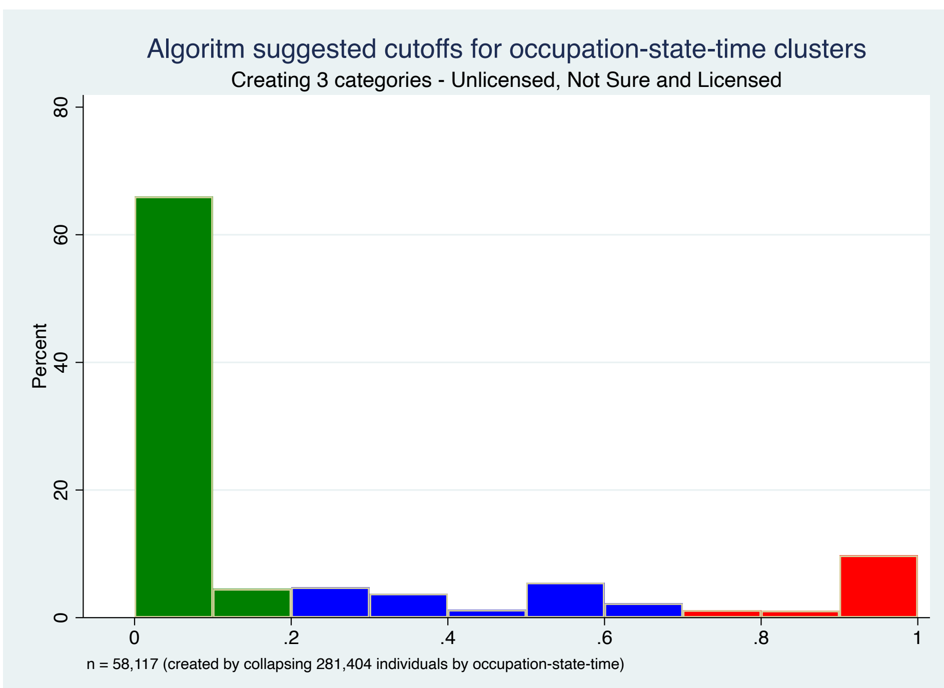
More formally, k-means clustering is an iterative procedure that divides the data into k groups or clusters. The procedure begins with k initial group centers, 2 in my case (for licensed and not licensed category), though my results are robust if I create 3 clusters (licensed, not licensed, or do not know category) too.

To implement the algorithm, we first assign all the observations to the group with the closest center. Second, the mean of the observations assigned to each of the groups is computed, and the first step is repeated again. These process continue until all observations remain in the same group from the previous iteration.

To avoid endless loops and attain algorithmic convergence, an observation is reassigned to a different group only if it is closer to the other group center. A tied distance between an observation and two or more group centers is resolved through assigning the observation to its current group if it is one of the closest, and to the lowest numbered group otherwise.

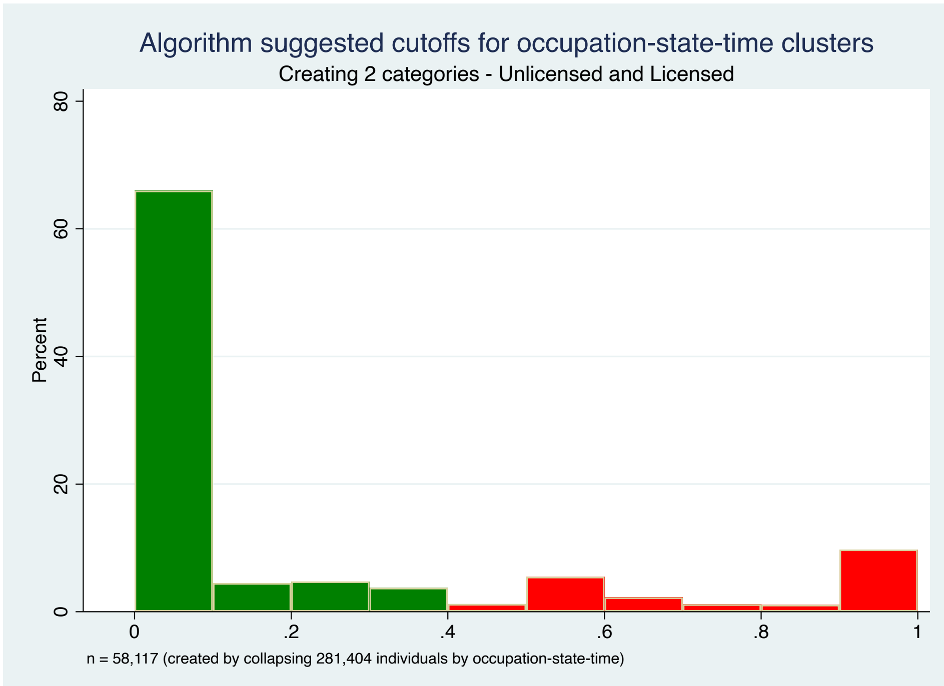
The two graphs, Figure 3 and Figure 4 below show how k-means generated cutoff points split the distribution of *proportion of people within-occupation-state-time clusters that are licensed* into 2 or 3 distinct categories. They show visually the decision rule of k-means clustering algorithm.

Figure 3: k-means algorithm with 3 clusters suggests that occupation-state-time clusters with average active government license response values below .219 are unlicensed, while occupation-state-time clusters with average active government license response values above .696 are licensed. The middle group is placed in a 'Not Sure' category, and we will drop these observations as we are not reasonably confident about their occupational license status due to noise.



Without this response correction, there is under-reporting of licensed status in licensed occupations, as discussed in the end of Section 3.3, and similarly over-reporting of unlicensed status in unlicensed states. In particular, looking at Figure 2, there might be systematic under-reporting of licensed status in licensed occupations over time, potentially due to three

Figure 4: k-means algorithm with 2 clusters suggests that occupation-state-time clusters with average active government license response values below .414 are unlicensed, while occupation-state-time clusters with average active government license response values above .416 are licensed.



different reasons. First, there could be some recall bias towards not being licensed if people are not well informed about the licensing status of their own occupation, maybe because they took their exam long time ago and have poor memory. Second, systematic under-reporting could also be due to some people who are actually in training and thus not certified but report the occupation in survey anyway. Third, systematic under-reporting error could come from occupations that are broad enough to include individuals who must be certified and others who do not need certification (even though I am using 4 digit occupation coding scheme based which is quite detailed already).

Similarly, for occupations that we know are not licensed by states, there may be some systematic over-reporting: some people within these occupation-states-time clusters may misreport that they are licensed when in fact they are not.

There are two reasons for making this correction. First, this k-means response correction will reduce measurement error, though it will not completely eliminate the attenuation bias as k-means algorithm itself may make some classification mistakes. Calculating accuracy (correctly detecting true positives and false negatives), specificity/recall (correctly detecting true positive) and precision (correctly detecting false negative) exercises can be undertaken for a small subset of occupations in a given year using the National Occupational Licensing Database (which provides information on whether 48 occupations are licensed or not in a state). Since these occupations were chosen by analysts on a certain criteria - high labor growth occupations licensed in 30 or more states and requiring less than a 4 year college education, the sample seemed unrepresentative and thus I did not yet calculate the three tests which can potentially help approximate/bound accuracy of k-means algorithm. Beyond these three measures, I am not aware of a statistical test which informs us of reduction in bias. Second, and the more important reason to use the k-mean response correction is to help us estimate an individual's last year occupation licensing status and thus test our preferred out-migration specification (see Section 3.4.2).

3.4.2 Changing Migration Variable Interpretation through Estimating Last Year Active Government License

Finally, after reducing the noise in active government license variable , I focus on the dependent migration variables.

Migrated variables in Annual Social and Economic Supplement (ASEC) indicates whether the respondent had changed residence in the past year. Non-movers, identified as individuals who were living in the same house as one year ago, were asked no more questions about migration over the past year. However, movers were asked asked where they lived last year in the US such as the city, county and state, or if they lived in a foreign country last year.

The category "Same house" includes both persons who did not move since the reference date (March 1 of the preceding year), and those who had moved but then returned to their earlier residence. On the other hand, movers are subdivided into the following 4 categories: those who had moved within the same county; those who had crossed county lines but stayed in the same state; those who those who had resided in a different state; and those who had migrated from abroad (Flood and Pacas 2018).

Appendix 8.1 provides more information on the original 280,404 individuals - Table 9 shows the summary statistics, Table 10 shows their current occupation frequency distribution and Table 11 shows their current state of residence frequency distribution.

Since our k-means corrected active government licensing variable tells us whether one is licensed or not by the state at the current moment, if I run a regression directly without any transformation, I would be answering is the following question - given that one is licensed by the state, did they move from another state/from abroad/from another county within state/from another house within county, or not move at all.

However, as I mentioned in Section 3.2, I prefer to test the out-migration specification (given that a person is licensed by the state, do they move out of the state,...) over the in-migration specification (given a person is licensed by the state, did they move into the state...).

Since I can observe the individual's last year occupation and state of residence last year in Annual Social and Economic Supplement, I just need to estimate if these individuals were licensed last year or not to be able to test my preferred specification, and there are two ways to do this.

First, I could use the same decision rule (cutoffs proportions generated by k-means cluster algorithm) to classify all last year occupations-state-time as licensed or unlicensed too, and thus estimate the license status of individual's last year occupation in their last year state of residence. For example, if in 2018 a person was a nurse last year and lived in CA last year, they would be licensed if more than 41.6% nurses in 2017 in CA were licensed, otherwise they are not licensed. If we were generating three clusters, the person in 2018 was licensed last year if more than 69.6% nurses in CA last year were licensed, and unlicensed if fewer than 21.9% nurses were licensed. Thus, this is a simple rule to estimate last year occupation licensing status.

Second, I could use a 'slight variant of the first simple rule' and use the decision rule cutoffs generated by k-means cluster algorithm only for people whose last year occupation and last year state of residence was in the data. For example, if in 2018 I find person was a nurse last year and lived in CA last year, I will only attribute them to be licensed or unlicensed last year if there are nurses in 2017 in CA in my merged CPS ASEC data (as the k-means algorithm was trained on this data). However, if there are no nurses in 2017 in CA in my merged CPS ASEC data, I cannot ascertain the license status of this person and will drop these individuals. Thus, this second method would allow me to have more confidence in my estimate of a person being licensed last year, but I may lose some individuals if there are no people within their last year occupation-state-time cluster (which could happen with small sized occupations).

I will use this second approach as it is more appropriate, and gives me similar results as the first simple approach. Thus, I transform the merged CPS ASEC data with corrected active government license variable for the 281,404 individuals one last time (this will be our master data going forward).

First, while every individual in our data has an occupation at time of survey, last year occupation is not known for some individuals. While there is no obvious pattern in the distribution of these 'Not in Universe' last year occupations across the years, occupations and other variables, almost 40% of these individuals are under 25, so presumably a lot of them were in school last year or not in the labor force last year. I drop these individuals whose last year occupations is not know (9,514 individuals).

Second, using the k-means generated cutoffs of 42% for 2 clusters and 22% and 70% for 3 clusters[21], I create estimates for last year active government license status for all individuals in the master data if their last year occupation-state-time cluster exists in the master data. However, I cannot estimate last year license status for another 71,453 individuals, and thus drop them too.

A major chunk of these 71,453 individuals whose last year license status cannot be estimated happens by design - because we use 'slight variant of the first simple rule'. Since my raw data starts from March 2016 onwards, I cannot estimate last year active government license for individuals in 2016 as there is no certification question data for March 2015 (jobcert question error in Current Population Survey). Thus, I loose 55,736 observations due to the design issue. Furthermore, I truly lose another 15,717 individuals over the other years because their last year occupation-state-time cluster does not actually exist in the master dataset, say because these are small sized occupations.

Thus, my last year active government license can only be estimated for 200,437 observations, down from 281,404 observations due to 9,514 missing last year occupation observations, and 71,453 individuals whose last year occupation-state-time cluster does not exist in my data (where all 55,736 individuals in 2016 are dropped).

I can however estimate last year active government license for people in March 2017 if there are people in their last year occupation-state-time (March 2016) cluster [22]. Also, note that I label individual's current active government license as 'licensed (end of year)' and label individual's last year active government license as 'licensed (beginning of year)' in summary statistics tables.

21. An additional 'not sure' category.

22. I do so by collapsing the master data by occupation-state-time cluster, and simply generating a new last year active government license variable that takes on a value of 1 if more than 42% people in the occupation-state-time cluster report they are licensed, and 0 otherwise (for the 2 cluster case). Similarly, I create a last year active government license variable with the 3 values (licensed, unlicensed and not sure). To merge these two variables back into the master dataset, I have to create a lagged time variable (last year) for all individuals in the master dataset, and rename the occupation-state-time clusters in the collapsed master data to last year occupation-state-time cluster. This allows me to do a m:1 match on the master data using the collapsed master data.

3.5 Summary Statistics

Table 4 shows the descriptive statistics of what I call the 'transformed data' that I use for preferred out-migration specification, in contrast to the original/non-transformed data in Appendix 8.1 that can be used for in-migration estimation (which is likely to suffer from endogenity).

Licensed (self-reported) binary variable takes on a value of 1 if the individual has an active government license. It was created from the three certification questions - profcert, statecert and jobcert, as shown in Table 3.

Table 3: Summary Statistics of Certification Questions (Transformed Data)

	Mean	Std. Dev.	Min	Max	Universe
Profcert: Has active prof-cert/license	0.254	0.44	0	1	200437
Statecert: Has govt issued prof-cert/license	0.913	0.28	0	1	50913
Jobcert: Cert required for job	0.845	0.36	0	1	50913
Licensed / Govt. issued prof-cert required for job	0.202	0.28	0	1	200437
Observations	200437				

Source: Current Population Survey (CPS), 2016-2020

Profcert Universe: 16-79 years old excluding people employed in Armed Forces.

Statecert Universe: Everyone who answered "yes" to Profcert. Those who answered "no" to Profcert are excluded.

Jobcert Universe: Everyone who answered "yes" to Profcert, excluding people "Not In the Labor Force" (NILF) and "Unemployed New Workers". Again, those who answered "no" to Profcert are excluded.

Licensed (self-reported) is added in the summary statistics to compare with our key parameter of interest - estimated active government license in last year, or equivalently active government license at beginning of year, as a sanity check. As already alluded, for the summary statistic table presentation, I will stick with the later interpretation/label for its relevance, and ease of contrast with non-transformed data. Also, note that licensed at beginning of year has fewer observations when it it is constructed from k-means 3 clusters than 2 clusters as I drop the individuals in "Not Sure" cluster when calculating the summary statistics for the variable. The mean of k-means corrected active government license at the beginning of current year reduces slightly compared to the active government license variable, but the standard deviation is a bit higher. Estimated active government license at beginning of year will help us test for geographic mobility of licensed workers, which may be low (better wages and job prospectus/regulatory capture theory) or high (more educated/ambitious workers)

Table 4: Summary Statistics

	Mean	Std. Dev.	Min	Max	Count
Licensed (self-reported)	0.202	0.28	0	1	200437
Licensed at beginning of year (k-means corrected with 2 clusters)	0.188	0.39	0	1	200437
Licensed at beginning of year (k-means corrected with 3 clusters)	0.150	0.36	0	1	157862
Proportion of people licensed in an occupation-year	0.188	0.31	0	1	200437
Migrated: Moved to a different state	0.017	0.13	0	1	200437
Migrated: Moved to a different county within state	0.022	0.15	0	1	200437
Migrated: Moved to another house within county	0.065	0.25	0	1	200437
Migrated: No / Stayed in Same House (Non-Mover)	0.896	0.31	0	1	200437
Home: Owned (or being bought)	0.694	0.46	0	1	200437
Home: Rented	0.306	0.46	0	1	200437
Age	43.393	14.31	16	79	200437
Gender: Male	0.517	0.50	0	1	200437
Race: White	0.816	0.39	0	1	200437
Race: Black	0.095	0.29	0	1	200437
Race: Other	0.089	0.28	0	1	200437
Educ: Has some schooling	0.073	0.26	0	1	200437
Educ: High School completed or equivalent	0.260	0.44	0	1	200437
Educ: Some college, Associate or Bachelor Degree	0.524	0.50	0	1	200437
Educ: Has a Master Degree, Professional or Doctorate Degree	0.143	0.35	0	1	200437
Marst: Living with partner	0.552	0.50	0	1	200437
Marst: No partner: Single/never married	0.288	0.45	0	1	200437
Marst: Not living with partner	0.160	0.37	0	1	200437
# of own children in household	0.813	1.13	0	9	200437
Total Personal Income (in 1000s)	62.628	79.63	-16.5	2085.7	200437
State of Residence at beginning of year			2	52	200437
Occupation at beginning of year			1	443	200437
Time Period			2016	2019	200437
Observations	200437				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020

Second, from the licensed at beginning of year variable, we can construct a new variable measuring the ‘proportion of people licensed in an occupation-year’. This variable will be a proxy for licensing costs on average that people in an occupation-year will have to pay if they move, and thus help us test for whether occupational licensing restricts geographic migration or not. The reported statistic in Table 4 is constructed by taking the ratio of number of **licensed** individuals (k-means corrected with 2 clusters) in an occupation in a year divided by number of total individuals in an occupation in a year. Thus, it is no surprise that the mean of licensed at beginning of year (k-means corrected with 2 clusters) is the same as mean of proportion of people licensed in an occupation-year, and the standard deviation of the later is lower.

Third, while we had 5 migration dummies in the non-transformed data which had a "moved from" migration interpretation, in the transformed data with a "moved to" interpretation of migration variable we should not theoretically have a "Migrated: Moved to abroad/different nation" category (compare Table 4 with Table 9 in Appendix 8.1). This happens because only people within the US are surveyed - asked from where did move from since last year/where they were living last year, and thus we know if a person lived abroad last year. However, if a person moves abroad, they are automatically dropped from the Current Population Survey, and thus move to abroad variable will have null values only for this migration category and no individual control data. As expected, the 4 migration "move to" dummy variables add exactly to 1 in Table 4. Also, it is interesting to note that almost 90% individuals did not move houses, while almost 2% moved to another state.

Fourth, we have 8 categories of individual level controls such as home ownership status, age, gender, race, education, marital status, number of own children residing in the household (of any age or marital status, irrespective of biological, adopted, or step-children), and total pre-tax personal income (from all sources for the last calendar year at the end of the year). These are observable at the end of year, which is not a major issue as most of these controls are largely time invariant (like race) or change at a relatively slow rate (like personal income). There is evidence though that home ownership and marital status may be important variables to influence migration decision for movers, so I will not include them in my preferred specification too as since they are measured at next year and could be affecting migration variable. My results are robust to including/excluding them though.

Fifth, occupations reports the person's primary occupation at the beginning of the year (one year ago). Respondents who held more than one job were to report the job at which they worked the largest number of hours. For persons who were employed at the time of the survey, occupation relates to the job worked during the preceding week; unemployed persons and those not currently in the labor force were to give their most recent occupation.

Occupation at beginning of year has 443 categories/occupations - one more than occupations at end of year, which is expected as the occupations at beginning of year are not restricted to occupations that have 10 or more people in an occupation at the end of the year (Section 3.4).

Furthermore, the unique number of categories of occupations at beginning of year/preceding year (shown in Table 4) does not differ much from unique categories of occupations at end of year/current year (shown in Table 9 in Appendix 8.1). Table 5 reports the frequency of occupations at the beginning of the year, and can be contrasted with Table 10 in Appendix 10 that shows frequency of occupations at end of year. Both these frequency tables also present the variation in occupational licensing across states in a randomly chosen year (2018). As expected, the number of observations within the unique occupation category at beginning of year/last year is lower than 10, unlike for occupations at end of year/current year which were restricted to have greater than 10 observations while setting up the k-means algorithm.

Table 5: Occupations Frequency Table (at beginning of year).
Proportion Licensed column shows proportion of people in the occupation that self-reported they require a license to practise their profession at the end of year.
Percent of licensed states in 2018 shows the share of total states that licensed the occupation in 2018 as per k-means algorithm, while total states in 2018 shows the count of all the states in which individuals practicing an occupation live at beginning of 2018.
As there are fewer people in an occupation, there may not be individuals in an occupation in a certain year, like 'Rail-Track Laying and Maintenance Equipment Operators'.

Occupation	Frequency	Percent of total obs - 200,437	Proportion Licensed (self reported)	Percent of States Licensed (k-means 2 corrected) in 2018	Total States in 2018
Managers, nec (including Postmasters)	7336	3.660	.145	0	51
First-Line Supervisors of Sales Workers	6452	3.219	.0743	0	51
Elementary and Middle School Teachers	5172	2.580	.712	94.1	51
Driver/Sales Workers and Truck Drivers	5123	2.556	.297	19.6	51
Registered Nurses	4913	2.451	.832	100	51
Retail Salespersons	4319	2.155	.029	0	51
Cashiers	4197	2.094	.0139	0	51
Secretaries and Administrative Assistants	3964	1.978	.0399	0	51
Chefs and Cooks	3537	1.765	.0535	0	51
Customer Service Representatives	3401	1.697	.0288	0	51
Janitors and Building Cleaners	3260	1.626	.0298	0	51
Waiters and Waitresses	2775	1.384	.0338	0	51
Laborers & Freight, Stock, & Material Movers, Hand	2744	1.369	.0285	0	51
Nursing, Psychiatric, and Home Health Aides	2721	1.358	.469	58.8	51
Construction Laborers	2634	1.314	.0654	1.96	51
Accountants and Auditors	2523	1.259	.221	7.84	51
Chief executives and legislators/public administration	2405	1.200	.165	11.8	51
Computer Scientists and Systems Analysts/Network systems Analysts/Web Developers	2328	1.161	.0608	0	49
...	...	...	...	...	
...	...	...	...	...	
...	...	...	...	...	
Paving, Surfacing, and Tamping Equipment Operators	1	0.000	0	0	1
Rail-Track Laying & Maintenance Equipment Operators	1	0.000	0		. .

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Sixth, states identifies the household's state of residence at the beginning of the year using the Federal Information Processing Standards (FIPS) coding scheme.

Similarly, state of residence at beginning of year has 51 unique categories/states (50 states plus D.C.) - it begins at 2 and ends at 52 because it has a placeholder category 1 for "Abroad", and there are no individuals in our transformed data set that were living abroad at the beginning of year. This again is a sanity check since we do not have certification data on those who were living abroad last year as they were not part of Current Population Survey, and thus k-means algorithm cannot estimate their last year licensing status using our 'slight variant of the first simple rule' (see Section 3.4.2). Thus, there are no people who moved from abroad in our out-migration specification.

The unique number of state of residence at the beginning of the year (Table 4) does not differ from unique number state of residence at the end the year. Table 6 reports the frequency of state of residence at beginning of year for the 10 largest and smallest state categories, and can be compared to largest and smallest occupation categories at end of year (Table 11 in Appendix 11).

4 Specification

Recall that we have a pooled cross-sectional data for individuals from 2016 to 2019. There may be some people who are in the sample twice due to CPS 4-8-4 rotation pattern.

My first hypothesis is that if a person works in a state licensed occupation, they are unlikely to migrate to another state over the next one year because of higher opportunity costs than unlicensed workers, partially due to higher current wages[23] and better career prospectus in licensed job (compared to unlicensed jobs in terms of tenure, employment status and other job advantages) (Nunn 2018). I call this '**cushy wage**' hypothesis.

My second hypothesis is that if an occupation is more likely licensed across all states in the country, individuals will not migrate to other states over the next one year from time t due to facing additional re-licensing costs (like monetary, psychic or time). I call this '**re-licensing cost**' hypothesis.

23. Just like occupational licensing literature suggests, I too find that k-means algorithm suggested licensed workers have higher incomes, thereby mitigating concerns about correctly identifying licensed individuals.

Table 6: State of Residence (at beginning of year) Frequency Table

State	Frequency	Total	Percent
California	18237	200437	9.099
Texas	11923	200437	5.949
Florida	8710	200437	4.346
New York	8378	200437	4.180
Illinois	5780	200437	2.884
Pennsylvania	5780	200437	2.884
Ohio	5377	200437	2.683
Massachusetts	4816	200437	2.403
Michigan	4527	200437	2.259
North Carolina	4490	200437	2.240
...	...	...	...
...	...	...	...
...	...	...	...
Iowa	2545	200437	1.270
Kansas	2536	200437	1.265
Nevada	2419	200437	1.207
South Dakota	2398	200437	1.196
Alaska	2247	200437	1.121
Kentucky	2193	200437	1.094
Connecticut	2002	200437	0.999
Delaware	1948	200437	0.972
Rhode Island	1655	200437	0.826
Maine	1574	200437	0.785

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

However, state occupational licensing should not affect individual migration within the state (to another county in same state, to another house in the same county, or not migrating at all/staying in the same house). Thus, the parameter coefficients for both these two hypothesis should be zero for within-state migration.

The model specification we have is Ordinary Least Squares (OLS) with a host of individual level controls, along with state, occupation, year, year-occupation and year-state fixed effects.

The estimating equations is the following, where i will index the individuals, t will index time, o indexes occupation and s indexes states -

$$\begin{aligned} Migration_{it} = \beta\ Licensed_{ost} + \delta\ Proportion\ Licensed_{ot} + \\ \gamma\ Controls_{it+1} + Occupation_o + State_s + Time_t \\ + Occupation_o * Time_t + State_s * Time_t + \epsilon_{it} \end{aligned} \tag{1}$$

$Migration_{it}$ is a dummy variable taking a value of 1 if a person moves to another state (location more generally) over the next one year from beginning of time period t.

$Licensed_{ost}$ is a dummy variable taking a value of 1 if a person is licensed by state s to practice occupation o at beginning of time period t, and measures the 'cushy wages' hypothesis. It was created from three certification questions and its misreporting has been reduced by applying the 2-mean cluster algorithm to the noisy self-reported measure.

$Proportion\ Licensed_{ot}$ measures percentage of people within an occupation that are licensed at time t (proxy for probability of re-licensing required for an occupation), and measures 'occuptional re-licensing costs' hypothesis.

$Controls_{it+1}$ include 5 categories of individual level control variables such as age, gender, race, education, and number of children under 5 years old in the household at end of time period t (or beginning of time period $t+1$). I have data on home ownership, marital status, and personal income at end of time period t too, which are reported in summary statistics Table 4, but I do not use them in my preferred out-migration specification as individuals tend to move due to first two factors and personal income may be a function of move, leading to over-controlling. Results stay robust to if I add these three controls though.

$State_s$ is state fixed effect term and controls for the (origination) state of residence at beginning of time period t. They are important because some states could have fundamentally different migration levels, say because of labor market conditions like unemployment or GDP, their political composition or even licensing propensity. Thus, I control for time invariant features of states.

$Occupation_o$ is the occupation fixed effect term and controls for individual's occupations at beginning of time period t. They are important because occupations can vary in terms of migration. For example, house builders are more likely to move across state borders for construction projects than say a massage therapist who may have a local clientele. We control for observable time invariant differences across occupations, and do not want a few occupations driving our results.

$Time_t$ is year fixed effect and controls fortime period t. These are important because there may be waves of more or less migration going on over time, maybe due to political think tanks trying to lobby for policy changes before upcoming elections, structural changes in jobs availability, etcetra. These affect every occupation and every state equally.

5 Results

Table 7 shows the key results.

Column 1 tests the hypothesis that given that a person is licensed in a state-occupation cluster at beginning of time period t, they would not move out to another state over the next one year as they would lose their 'cushy' licensed job benefits. Columns 2-4 serve as falsification tests, and the outcome variable changes to migration to another county within state, migration to another house within county, and to no migration (non-movers). Migration patterns within the state should not be adversely affected as individuals do not lose their state-license as long as they stay in the same occupations[24].

In column 1, we can see that if a person is currently working in a (state) licensed occupation, they are more likely to move to another state over the next one year by .11 percentage point. Since only about 1.7 percent people migrate to another states every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 6.5% more than non-licensed individuals on average, cetrius paribus. While this seems to be moderate in magnitude, this is not a statistically significant result even at 10% level of significance. I do not find any evidence that licensed individuals are moving less than across state borders due to losing their 'cushy' jobs.

Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .41 percentage points compared to individuals in a unlicensed occupation in all states in a given year, ceteris paribus. Again, this is not statistically significant, and I

24. Robustness check where I limit to individuals whose last year and current year occupation was the same yields similar results.

Table 7: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0011	0.0008	-0.0013	-0.0005
	(0.0012)	(0.0014)	(0.0023)	(0.0028)
Proportion Licensed (prob. of re-licensing costs)	-0.0041	-0.0053	0.0055	0.0039
	(0.0068)	(0.0079)	(0.0127)	(0.0158)
Age	-0.0013***	-0.0017***	-0.0057***	0.0086***
	(0.0001)	(0.0002)	(0.0003)	(0.0003)
Age Squared	0.0000***	0.0000***	0.0000***	-0.0001***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Gender: Male	-0.0010	-0.0010	-0.0019	0.0039**
	(0.0007)	(0.0008)	(0.0014)	(0.0017)
Race: White	-0.0017	-0.0011	-0.0031	0.0058**
	(0.0013)	(0.0013)	(0.0022)	(0.0027)
Race: Black	0.0014	0.0009	0.0113***	-0.0136***
	(0.0016)	(0.0017)	(0.0029)	(0.0036)
Educ: High School completed or equivalent	0.0056***	0.0046***	0.0164***	-0.0266***
	(0.0011)	(0.0014)	(0.0024)	(0.0029)
Educ: Some college, Associate or Bachelor Degree	0.0085***	0.0076***	0.0134***	-0.0294***
	(0.0011)	(0.0014)	(0.0024)	(0.0029)
Educ: Masters, Professional or Doctorate Degree	0.0132***	0.0071***	0.0168***	-0.0371***
	(0.0015)	(0.0017)	(0.0029)	(0.0036)
# of own children in household	-0.0037***	-0.0027***	-0.0054***	0.0118***
	(0.0003)	(0.0003)	(0.0005)	(0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
r2	.014739	.0115908	.0316533	.0464488
N	200,429	200,429	200,429	200,429

Dependent variable (migration) is a dummy. Constant term is not reported.
* $p < .10$, ** $p < .05$, *** $p < .01$

do not find evidence for 'occupational re-licensing costs' reducing inter-state migration.

The coefficients on other parameters in column 1 are as expected though. One is less likely to migrate to another state as they get older (the positive quadratic age co-efficient is much smaller in magnitude than the negative linear age co-efficient), but once people are "old" enough they are more likely to migrate out of their licensed occupation-state. This suggests that maybe as people build local reputation or clientele, they are less likely to move out until they decide to retire. Also, people with more education are likely to move to another state than compared to people with only some schooling, and magnitudes are higher for people with more formal education, suggesting that people with more more ambition/skills are more likely to make inter-state moves. Furthermore, the more the number of own children in household, the less likely is one to leave the state, presumably because reallocation costs will be higher as family size increases.

In Appendix 8.2, Table 16 shows how the key parameters (licensed and proportion licensed) change as I add the 5 control variables one by one, and Table 17 shows how the key parameters change as I add fixed effects one by one. In the later table, I also add occupation-time interaction fixed effects term to controls for trends in occupations over the time period t[25] and state-time interaction fixed effects term to controls for trends in state over time period t[26]. I usually avoid adding state-occupation interaction terms in my baseline specification as we have corrected our active government license measure using k-means algorithm so that there is no variation in individual responses within occupation-state-time clusters, and after including the three plain vanilla fixed effects, adjusted R-square does not increase much[27] as I add the interaction fixed effects but the interacted fixed effects often results in one of our key parameters becoming infeasible to estimate.

Also, from Table 17 in Appendix 8.2, the first three columns show proportion licensed variable stays significant when we add time/year fixed effects alone, or state fixed effects alone, but the parameter looses significance as soon as we add occupation fixed effects. This suggests that there seems to be much more variation across occupations than states, and

25. Occupation-time fixed effects may be important because some occupations like truck driving may have been lucrative in 2016, but not so much in 2019 as they may have got saturated (excessive competition). These people may be migrating to other states in search of opportunities. If this co-incides with introduction of occupational laws, it will bias our active government licensing results downward.

26. State-time fixed effects may be important because states like New York or California may have been a great place to live in 2016, but a not so great in 2019, say due to cost of living or medical infrastructure being less available in these states. People may not have wanted to move out in 2016, but may want to move out in 2019. If some occupations in New York or California coincided with introduction of introduction of occupational licensing laws, it will bias our active government licensing variable downwards.

27. In column 7, R-square is 1.47% while adjusted R-square is 1.23%, and in column 9 R-square is 2.24% while adjusted R-square is 1.37%

least variation across time. One potential reason there is positive and statistically significant coefficient on proportion licensed without occupation fixed effects, but not with occupation fixed effects, is that licensing is more appealing in occupations in which mobility is already low. Alternatively, states seem to be more consistent in their licensing laws, but there is quite a lot of variation in licensing by occupations that gets absorbed by the occupation fixed effects.

In column 2 (of Table 7), we can see that if a person is currently working in a (state) licensed occupation, they are more likely to move to another county within the state over the next one year by .08 percentage point. Since about 2.2 percent people migrate to another county within state every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 3.64% more than non-licensed individuals on average, cetrius paribus. This is neither large in magnitude nor statistically significant at 10% level of significance, and coefficients on controls variables follow the same pattern like in column 1 and are about similar in magnitudes too. I would expect the opportunity cost of licensed individuals is lower for within-state moves, but again find they are not moving significantly more between counties within the state. Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .53 percentage points compared to individuals in a unlicensed occupation in all states in a given year, ceteris paribus. Again, this is not statistically significant, and I do not find evidence for 'occupational re-licensing costs' reducing intra-state county migration, as expected.

In column 3, we can see that if a person is currently working in a (state) licensed occupation, they are less likely to move to another house within county over the next one year by .13 percentage point. Since about 6.5 percent people migrate to another county within state every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 0.02% less than non-licensed individuals on average, cetrius paribus. Again, this is neither large in magnitude and nor statistically significant at 10% level of significance, and coefficients on controls variables follow the same pattern like in column 1 and are about similar in magnitudes too. Furthermore, since opportunity cost of licensed individuals is lower for within-state moves, I would expect them to move more within county in a state but do not find any statistical significant result. Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .55 percentage points compared to individuals in a unlicensed occupation in all states in a given year, ceteris paribus. Again, this is not statistically significant, and I do not find evidence for 'occupational re-licensing costs' reducing intra-county house migration.

Finally, in column 4, we see that if a person is currently licensed, they are more likely to stay in the same house (within state) over the next one year by .05 percentage point i.e. not migrate. Since about 89.6 percent people stay in the same house on average, individuals in state licensed occupations seems to be staying in the same house by about 0.005% more than non-licensed individuals on average, cetrius paribus. Again, the magnitude of this estimate is very small, and the result is not statistically significant, suggesting some evidence for 'cushy wage hypothesis'. Additionally, individuals in a licensed occupation in all states are more likely to not move to another place .39 percentage points compared to individuals in a unlicensed occupation in all states in a given year, ceteris paribus. Again, this is not statistically significant, and I do not find evidence for 'occupational re-licensing costs' reducing inter-state migration.

The signs on the certain control variables should flip, as the outcome variable is measuring if the person did not move now. This makes the story we have painted with the first three columns a bit more consistent, by construction. For example, as a person gets older or has more number of children in household, they are more likely to stay in the same house. Furthermore, if a person is more educated, they are less likely to stay in their own home. Also, it is interesting to note that individuals belonging to black race seem to be less likely to stay in the same household (as compared to all other races), and more likely to move to another house within the county in a year.

5.1 Robustness Check: k-means algorithm identifies licensed individuals

To test whether k-means algorithm correctly identifies licensed occupation-state time or not at a given moment of time, I can test for whether licensed individuals (as per k-means algorithm) have higher incomes or not, like occupational licensing literature argues and usually finds due to regulatory capture. Finding licensed individuals have higher wages would add confidence that my results are not contaminated by inability of the k-means algorithm to correctly identify licensed individuals, and I will use the non-transformed/original data (281,404 observations in Table 9, Appendix 8.1) for this.

The estimating equations is the following, where i will index the individuals, t' will index time (end of year), o' indexes occupation (at end of year) and s' indexes state (at end of year) -

$$\begin{aligned} Personal\ Income_{it'} = \beta\ Licensed_{o's't'} + \gamma\ Controls_{it'} + Occupation_{o'} + State_{s'} + Time_{t'} \\ + Occupation_{o'} * Time_{t'} + State_{s'} * Time_{t'} + \epsilon_{it'} \end{aligned} \tag{2}$$

$Personal\ Income_{it'}$ is the individual's total pre-tax personal income (or losses) from all sources for the last one calendar year at time t'. $Licensed_{o's't'}$ measures if a person is licensed or not in occupation o' in state s' at time t', constructed from the three certification questions and corrected for mis-reporting using k-means algorithm. $Controls_{it'}$ are defined at the same time period (end of year) when income is measured, unlike in estimating Equation 1, and additionally include home ownership and marital status on top of age, gender, race, education attainment and number of own children in household. $Time_{t'}$ controls for year fixed effects. $Occupation_{o'}$ and $State_{s'}$ are fixed effects controlling for individual's current occupation and state of residence at end of year t', and $Occupation_{o'} * State_{s'}$ and $State_{s'} * Time_{t'}$ are interaction fixed effect terms.

Table 12 in Appendix Section 8.1.1 shows the regression results. Licensed individuals make about \$2,000 dollars more than unlicensed individuals as per k-means 2-cluster algorithm, and about \$4,000 dollars more than unlicensed individuals as per k-means 3-cluster algorithm. Coefficients on other parameters also make sense, like income increases by approximately \$1,000 dollars every year with age/experience/human capital, males makes about \$20,000 more females, white make about \$3,000 more while blacks make about \$1,000 less than other races, and individuals with advanced degrees (masters, profession or doctorate) make about \$40,000 dollars more than those with some high school degree, cetrius paribus.

Similarly, if I replace the dependent variable in equation 2 with employment status of workers, I find some some preliminary evidence that licensed individuals are half to one percentage point less likely to be unemployed than their unlicensed counterparts (Table 13 in Appendix Section 8.1.2).

$$\begin{aligned} Unemployed\ Worker_{it'} = \beta\ Licensed_{o's't'} + \gamma\ Controls_{it'} + Occupation_{o'} + State_{s'} + Time_{t'} \\ + Occupation_{o'} * Time_{t'} + State_{s'} * Time_{t'} + \epsilon_{it'} \end{aligned} \tag{3}$$

where $Unemployed\ Worker_{it'}$ takes on a value of 1 if the individual is unemployed, and $Controls_{it'}$ includes all variables in equation 3 and total personal income.

Furthermore, if I replace the dependent variable in equation 2 with log of total number of people within an occupation-state-time cluster and collapse my individuals level data by occupation-state-time cluster, I find some preliminary evidence that if an occupation-state-time cluster is licensed, there would be 5 to 14% fewer workers in them (Table 14 in Appendix

Section 8.1.3).

$$
\begin{aligned}
LogN_{o's't'} = \beta \ Licensed_{o's't'} + \gamma \ Controls_{o's't'} + Occupation_{o'} + State_{s'} + Time_{t'} \\
+Occupation_{o'} * Time_{t'} + State_{s'} * Time_{t'} + \epsilon_{it'}
\end{aligned} \tag{4}
$$

This suggests that regulatory capture effect that reduces the number of workers in licensed occupation dominates the adverse selection effect that increases the number of workers in licensed occupations.

5.2 Robustness Check: Alternative Specifications

The results are robust to alternative specifications. First, even when I use the corrected active government license variable with three clusters instead of two, I get similar results (see Table 20 for summary statistics and Table 21 for regression results in Appendix 8.3). Furthermore, the results are robust to using different filters on training data to determine the k-means algorithm decision rule, such as dropping occupations with less than 20 people or 50 people instead of occupations with 10 people only.

Second, adding housing, marital status and total personal income controls in the k-means 2 or 3 cluster approach does not not change the results either (over controlling). Table 18 in Appendix Section 8.2.1 shows the results for k-means 2 cluster.

Third, if I relax the definition of occupational licensing to the state license definition i.e. "yes" to statecert question only, and not "yes" to both statecert and jobcert question (which is what I used/stronger definition), the results do not change as the active government license variable stays almost the the same (results not attached). Similarly, if I relax the definition of occupational licensing from someone who answered "yes" to all the three certification questions (primary job is licensed) to someone who just answered "yes" to "profcert" question (holds an active license of certification) only, my results do not change.

Fourth, if I change the sample of individuals from 16 to 79 in my data to a working age sample of 25 to 54, it does not affect my results (results not attached).

Fifth, if I restrict myself to continuing workers only, defined as those whose last year occupation was the same as current year occupation (as opposed to new workers), I lose about 10% of observations but my results do not change (Table 19 in Section 8.2.2). This is an important result, as it allows us to rule out a particular explanation to why we get the results. Consider a toy model where there there are two types of workers - licensed and unlicensed who initially have similar wages and number of employed people. As licensing gains momentum (like it has in the US), labor supply of workers in licensed occupations shifts to the left (regulatory capture effects) and thus wages of licensed individuals will increase while the number of people employed in licensed jobs decreases (I find evidence for both these effects). Thus, while there are fewer people in licensed occupations, some of the individuals who lost their jobs due to regulatory capture could move to different states to pursue the same occupation, and this effect could potentially explain why I do not find evidence for re-licensing costs hypothesis. However, by focusing only on people who stay in the same occupation over the year, we can rule out this possible explanation to be driving my empirical results.

Sixth, weighting the regression by number of people in the occupation-state-time cluster also does not change my results either (results not attached).

Finally, I change the preferred out-migration specification (equation 1) to an in-migration specification and test for the following hypothesis - given that an individual is currently self-reported licensed by the state in an occupation at end of year t, did the individual move into the state-occupation over the last one year? I find some evidence that licensed individuals are less likely to have migrated from another state or abroad (Table 15 in Section 8.1.4) suggesting there are licensing costs too, but the in-migration specification is likely to be endogenous given self-selection of individuals into unlicensed occupations while migrating to US state.

5.3 Making sense of the results

Table 8: Reason for Moving (asked from movers only)

Reason for moving	Frequency	Total	Percent	Proportion Licensed
Housing Related Reason	8347	20840	40.1	0.1675
Family Reason	6002	20840	28.8	0.1671
Job Related Reason	4542	20840	21.8	0.1808
Other Reasons	1949	20840	9.4	0.1380

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020.
Non-movers are not asked why they moved.

The results can be partially explained by examining individual's reasons for moving since movers are asked why they moved. In our dataset of 200,437 individuals, about 89.6% people did not move, suggesting moving costs may be high, and thus only the remaining roughly 10.4% of 200,437≈ 20,840 people were asked this question.

Table 8 shows that about 40% people migrate for housing related reasons. This includes people who wanted to own home (not rent), wanted new or better housing, wanted better neighborhood, and moved for cheaper housing amongst other housing reasons. The high magnitude on the home ownership (as opposed to renting home) as well as statistical significance at 1% level of significance seems to suggest that people do not move if they own a home, and seem to be moving to another state to own a home.

About 30% people move due to family related reasons, which include change in marital status, to establish their own household, and pursuing a relationship with unmarried partner amongst other family reason.

Only 20% people move for job related reasons, which includes a new job or job transfer, to look for work or lost job, for easier commute, and retirement amongst other job-related reason. Thus, it appears that migrating for work related reasons is not the highest priority for many individuals.

The remaining 9% people moved due to other reasons such as attending or leaving college (2.5%), health related reason (.9%), Change of Climate and Natural Disaster (.8%), and foreclosure/evictions.

Thus, it seems like people do not usually move for job related reasons, while other factors like owning a home or starting a family are more important reasons to move. Furthermore, from the geographic mobility literature, we know that moving is costly because of search costs, transitions/direct moving costs, and possible upheaval or psychic costs. In the presence of these costs, maybe occupational licensing is not the most important component.

6 Conclusion

6.1 Method

In summary, I use three recently added non-degree credential questions in Current Population Survey from 2016 onwards[28] to estimate whether a person is licensed by the state government[29] to work in their primary occupation or not. The individual-level active government license dummy estimate based on responses to the three certification questions seems to suffer from misreporting[30], and I correct the misreporting by standardizing the responses

28. These three questions were - do you hold an active professional certificate or license (profcert), was the active professional certification or license issued by local, state or federal government (statecert), and whether this active professional certificate or license was required for your primary job (jobcert).

29. Since Federal Trade Communication Technical Report in 2018 affirms that "occupational licensing is almost always state-based", if a person answers "yes" to the three certification questions we can infer they are likely licensed by the state government (Goldman 2018).

30. Within an occupation-state-time cluster, every individual should give the same uniform replies - either licensed or unlicensed. In particular, if some people report they are licensed while others report they are not

of people within each occupation-state-time cluster using k-means unsupervised machine learning algorithm.

In particular, k-means clustering algorithm with 2 categories suggests that all individuals within occupation-state-time cluster where more than 42% people self-report holding a license should be licensed, while everyone else should be unlicensed. Alternatively, if one was to create 3 categories, the algorithm suggests that all individuals within occupation-state-time cluster where more than 69% people self-report to be licensed are likely licensed, while all individuals within occupation-state-time clusters where less than 21% people self-report to be licensed are likely unlicensed - and we cannot be sure of everyone else. Second, k-means algorithm allows me to estimate an individual's last year occupational licensing status, and thereby test my preferred out-migration specification instead of the default in-migration specification, based on data collected in CPS and ASEC.

6.2 Results

Like both theoretical and empirical literature suggests, I too find that occupational licensing creates a wage gap/inequality between licensed and unlicensed workers - thereby adding more credibility to my k-means algorithm decision rule that is correctly classifying occupation-states as licensed and unlicensed at any given moment of time.

I find that between state migration patterns of individuals are not affected if an individual currently works in a (state) licensed occupations. However, I do find that within state migration patterns of individuals are not affected if an individual currently works in a (state) licensed occupations, as one would expect. Thus, (state) occupational licensing does not seem to adversely affect geographic mobility for licensed professionals, and cannot help explain the stylized US labor facts.

Second, I do not find much evidence for occupational relicensing costs too, just like I did not find support for cushy wages hypothesis. This could be simply because migration decision is very complex and there are major factors like home ownership or family reasons that trump job related reasons to migrate. We saw that only 20% movers moved for job related reasons, which is relatively lower compared to other factors like housing (40%) or family related reasons (30%).

My preferred out-migration specification results are quite robust to a wide range of specifications including choice of demographic controls or sub-sample analysis based on age, continuing/new workers, construction of number of clusters by k-means algorithm (2 or

suggests misreporting.

3), choice of occupations-state to determine the k-means cutoff rule, weighting schemes, et cetra.

6.3 Future Work/Potential Extentions

While I find that licensed individuals are no different than their unlicensed counterparts on average in terms of their migration choices, and occupational licensing costs do not seem to be restricting inter-state geographic mobility of American workers, occupational licensing could be adversely impacting other margins of outcome like wage inequality, potential employment and employment status (for which I provide preliminary evidence). Additionally, occupational licensing could also be affecting quality/safety of products and/or product prices that could be future study topics.

Also, the study can be improved by controlling for interstate occupation reciprocity agreements that have gained popularity in a few professions like nursing that I do not currently account for.

Furthermore, I intend to improve Kliener and Johnson's (2020) causal estimates of occupational licensing on inter-state migration by using the k-means suggested licensed occupations in Current Population Survey (CPS) to create a mapping of occupations in American Community Survey (ACS) data so that their re-licensing costs can be more precisely estimated. Currently, their licensing costs are within my 95% confidence interval, so my work does not necessarily contradict their study.

6.4 Contribution

My study looks at the impact of occupational licensing on geographic mobility of workers in pretty much the entire occupation set in US, unlike most other occupational licensing studies that focus on a single profession only and only try to see how licensing affects geographic mobility in a certain occupation. Simultaneous licensing of multiple occupations is closer to a general equilibrium effect whereas single occupation licensing effects is only a partial equilibrium effect. Furthermore, since almost one in four Americans are licensed today, this is an important question, especially if one believes in labor mobility is required for smooth functioning of the economy.

As far as I know, this is the first study that exploits the three certification questions in Current Population Survey to study the impact of occupational licensing on geographic mobility. of workers. Unlike Kliner and Johnson (2010), who also look at the same question - causal impact of (state) occupational licensing on inter-state mobility, I do not assume away time and geographic variation in occupational licensing or focus on only a small subset of occupations. In particular, they did not exploit the certification questions in CPS and rely more on ACS data.

Unlike Nunn (2016) who did use the Current Population Survey certification questions to study migration behavior of licensed individuals in 2016, my results are not driven by unobservable differences between states and occupations as I account for these using fixed effects (not accounting for these explains his results), and I also use a stronger set of time varying controls. Furthermore, I even correct for misreporting in individual licensing status that can potentially drive geographic mobility results through the k-means unsupervised machine learning algorithm.

The results have significance for public policy makers trying to understand worker effect of labor certification process on worker decisions about geographic mobility. As occupational licensing gets more and more popular within political circles over time and space, understanding its impact on workers can help politicians make more informed policy decisions. Also, since this study suggests that licensed workers seems to get positively affected on their wage and employment margin, policy makers have to weigh the freedom granted to licensed occupations to move between regions versus the negative externalities on non-licensed professionals in terms of wages and employment outcomes. Even if policymakers intend on simply increasing product quality or safety in product markets, they should consider evidence from this paper on worker outcomes, including in terms of mobility.

7 Bibliography

Allard, Mary Dorinda. “Adding questions on certifications and licenses to the Current Population Survey.” Monthly Labor Review, U.S. Bureau of Labor Statistics, November 2016.

Arrow, Kenneth. “The Theory of Discrimination.” Princeton University, Department of Economics, Industrial Relations Section. Working Papers 403 (1971).

Borjas, George J. “Does Immigration Grease the Wheels of the Labor Market.” Brookings Papers on Economic Activity I (2001): 69–133.

Brian Isom, Jacob M. Caldwell. Does Occupational Licensing Impact Access to Dental Care? Technical report 004. The Center for Growth and Opportunity at Utah State University, August 2019.

Carpenter, Dick M. II. “Testing the Utility of Licensing: Evidence from a Field Experiment on Occupational Regulation.” Journal of Applied Business and Economics 13, no. 2 (2012): 28–41.

DePasquale, Christina, and Kevin Stange. “Labor Supply Effects of Occupational Regulation: Evidence from the Nursing Licensure Compact.” NBER Working Paper Series, no. 22344 (2016).

Flood, Sarah M., and Pacas,José. “Using the Annual Social and Economic Supplement as Part of a Current Population Survey Panel.” Journal of Economic and Social Measurement 42, nos. 3-4 (May 2018): 225–248.

Follett, Tatiana, Iris Hentze, and Zach Herman. Universal Licensure Recognition. Technical report. National Conference of State Legislatures, March 2021.

Goldman, Karen A. Policy Perspectives: Options to Enhance Occupational License Portability. Technical report. Federal Trade Commission, September 2018.

Goodwin, James S., Figaro L. Jr. Loresto, Linda R. Rounds, and Yong-Fang Kuo. “States With The Least Restrictive Regulations Experienced The Largest Increase In Patients Seen By Nurse Practitioners.” Health Affairs 7 (July 2013): 1236–1243.

Hentze, Iris. Occupational Licensing and Military Spouses: States Consider New Solutions. Technical report. National Conference of State Legislatures, March 2021.

Hugh Cassidy, Tennecia Dacass. “Occupational Licensing and Immigrants.” The Center for

Growth and Opportunity at Utah State University i, no. Working Paper 2019.009 (July 2019).

Janna E. Johnson, Morris M. Kleiner. “Is Occupational Licensing a Barrier to Interstate Migration?” Research Division, Federal Reserve Bank of Minneapolis, no. Staff Report 561 (December 2017).

Johnson, Janna E., and Morris M. Kleiner. “Is Occupational Licensing a Barrier to Interstate Migration?” American Economic Journal: Economic Policy 12, no. 3 (August 2020): 347–373.

Kleiner, Morris M., and Alan B. Krueger. “The Prevalence and Effects of Occupational Licensing.” British Journal of Industrial Organisation 0007, no. 1080 (2010).

Kleiner, Morris M., and Krueger, Alan B.. “The Prevalence and Effects of Occupational Licensing.”British Journal of Industrial Relations 48:4, nos. 0007–1080 (December 2010): 676–687.

Kleiner, Morris M. and Todd, Richard M.. “Mortgage Broker Regulations That Matter: Analyzing Earnings, Employment, and Outcomes for Consumers.” National Bureau of Economic Research, no. Working Paper 13684 (December 2007).

Kudrle, Morris M., and Kleiner Robert T. “Does Regulation Affect Economic Outcomes? The Case of Dentistry.” The Journal of Law and Economics 43, no. 2 (October 2000): 547–582.

Liang, J. Nellie, and Jonathan Ogur. Restrictions on Dental Auxiliaries: An Economic Policy Analysis. Technical report. Bureau of Economics Staff Report to Federal Trade Commission, May 1987.

Marier, Allison, Morris M. Kleiner, Kyoung Won Park, and Coady Wing. “Relaxing Occupational Licensing Requirements: Analyzing Wages and Prices for a Medical Service.” National Bureau of Economic Research, no. Working Paper 19906 (February 2014).

Maya N. Federman, Kathy J. Krynski, David E. Harrington. “The Impact of State Licensing Regulations on Low-Skilled Immigrants: The Case of Vietnamese Manicurists.” The American Economic Review 96, no. 2 (May 2006): 237–241.

Molloy, Raven, Christopher L. Smith, and Abigail Wozniac. “Declining Migration in US: The Role of the Labor Markets.” NBER Working Paper Series, no. 20065 (2014).

Molloy, Raven, Christopher L. Smith, and Abigail Wozniak. “Internal Migration in the United States.” Journal of Economic Perspectives 25, no. 3 (Summer 2011): 173–196.

Mulholland, Sean E., and Young, Andrew T.. "Occupational Licensing and Interstate Migration." Cato Jounal 36, no. 1 (Winter 2016): 17–32.

Nunn, Ryan. How Occupational Licensing Matters for Wages and Careers. Technical report. The Brookings Institution, March 2018.

Nunn, Ryan. Occupational Licensing and American Workers. Technical report. The Brookings Institution, April 2016.

Pashigan, B. Peter. "Occupational Licensing and the Interstate Mobility of Professionals." NBER Working Paper Series, no. 220 (December 1977).

Sanders, Jon. New report on occupational licensing makes a strong case for reform. Technical report. John Locke Foundation, March 2020.

Shierholz, Heidi. The number of workers represented by a union held steady in 2019, while union membership fell. Technical report. Economic Policy Institue, January 2020.

The Evolving State of Occupational Licensing: Research, State Policies and Trends, 2nd Edition. Technical report. National Conference of State Legislatures, November 2019.

Thomas J. Kane, Douglas O. Staiger, Jonah E. Rockoff. "What does certification tell us about teacher effectiveness? Evidence from New York City." Economics of Education Review 27 (2008): 615–631.

Thornton, Robert J., and Edward J. Timmons. "The Delicensing of Occupations in the United States." Monthly Labor Review, U.S. Bureau of Labor Statistics, May 2015.

Treasury Office of Economic Policy, Department of the, The Council of Economic Advisers, and Department of Labor. "Occupational Licensing: A Framework for Policy Makers," July 2015.

U.S. Bureau of Labor Statistics. "Frequently Asked Questions about Data on Certifications and Licenses." Labor Force Statistics from the Current Population Survey, April 2017.

8 Appendix

8.1 Summary Statistics of Non-Transformed Data

Table 9: Summary Statistics

	Mean	Std. Dev.	Min	Max	Count
Licensed (self-reported)	0.196	0.40	0	1	281404
Licensed at end of year (k-means corrected with 3 clusters)	0.121	0.33	0	1	224718
Licensed at end of year (k-means corrected with 2 clusters)	0.184	0.39	0	1	281404
Migrated: Moved from a different state	0.018	0.13	0	1	281404
Migrated: Moved from abroad / different nation	0.003	0.05	0	1	281404
Migrated: Moved from a different county within state	0.023	0.15	0	1	281404
Migrated: Moved within county in the same state	0.067	0.25	0	1	281404
Migrated: No / Stayed in Same House (Non-Mover)	0.889	0.31	0	1	281404
Home: Owned (or being bought)	0.688	0.46	0	1	281404
Home: Rented	0.312	0.46	0	1	281404
Age	43.076	14.41	16	79	281404
Gender: Male	0.521	0.50	0	1	281404
Gender: Female	0.479	0.50	0	1	281404
Race: White	0.815	0.39	0	1	281404
Race: Black	0.097	0.30	0	1	281404
Race: Other	0.088	0.28	0	1	281404
Educ: Has some schooling	0.079	0.27	0	1	281404
Educ: High School completed or equivalent	0.264	0.44	0	1	281404
Educ: Some college, Associate or Bachelor Degree	0.520	0.50	0	1	281404
Educ: Has a Master Degree, Professional or Doctorate Degree	0.138	0.34	0	1	281404
Marst: Living with partner	0.544	0.50	0	1	281404
Marst: No partner: Single/never married	0.296	0.46	0	1	281404
Marst: Not living with partner	0.161	0.37	0	1	281404
# of own children in household	0.807	1.13	0	9	281404
Total Personal Income (in 1000s)	59.149	76.50	-16.5	2085.7	281404
State of Residence at end of year			1	51	281404
Occupation at end of year (ASEC)			1	442	281404
Survey year			2016	2020	281404
Observations	281404				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020

Licensed binary variable takes on a value of 1 if the individual requires a govt. issued professional certificate for their primary job, and 0 otherwise.

Note that almost 89% individuals did not move, while slightly less than 2% moved from a different state. Even fewer (.3%) moved to US from abroad, and note that if somebody was to leave the US, I would not be able to observe where they migrated to (as they would not be included in Current Population Survey anymore).

Occupations reports the person's primary current occupation. If the individual was employed during the survey week, their occupation for the last week was recorded, while if he/she were not in the labor force or employed, their last occupation was recorded. Furthermore, if the individual held multiple jobs, the job where they worked the longest hours was recorded. **Table 10 below** reports the frequency of occupations for the 23 largest and smallest occupation categories.

States identifies the household's current state of residence using the Federal Information Processing Standards (FIPS) coding scheme. Table 11 below reports the frequency table of state of current residence.

Table 10: Occupations Frequency Table (at end of year).
Proportion Licensed column shows proportion of people in the occupation that self-reported they require a license to practise their profession.
Percent of licensed states in 2018 shows the share of total states that licensed the occupation in 2018, while number of states in 2018 shows the count of all the states in which individuals practising an occupation live.

Occupation	Frequency	Percent of total obs - 281,404	Proportion Licensed (self reported)	Percent of States Licensed (k-mean2 corrected) in 2018	Total States in 2018
Managers, nec (including Postmasters)	8998	3.198	.142	0	51
First-Line Supervisors of Sales Workers	8163	2.901	.0757	0	51
Driver/Sales Workers and Truck Drivers	6541	2.324	.292	11.8	51
Elementary and Middle School Teachers	6530	2.321	.71	98	51
Registered Nurses	6212	2.208	.825	100	51
Cashiers	5772	2.051	.0125	0	51
Retail Salespersons	5607	1.993	.0294	0	51
Secretaries and Administrative Assistants	5071	1.802	.039	0	51
Chefs and Cooks	4710	1.674	.0512	0	51
Janitors and Building Cleaners	4361	1.550	.0268	0	51
Customer Service Representatives	4329	1.538	.0275	0	51
Laborers and Freight, Stock, & Material Movers, Hand	3616	1.285	.0271	0	50
Nursing, Psychiatric, and Home Health Aides	3574	1.270	.456	60.8	51
Waiters and Waitresses	3478	1.236	.0348	0	51
Construction Laborers	3368	1.197	.0591	0	51
Accountants and Auditors	3255	1.157	.22	7.84	51
Chief executives and legislators/public administration	3025	1.075	.16	9.8	51
Computer Scientists and Systems Analysts	2949	1.048	.0627	1.96	51
Stock Clerks and Order Fillers	2709	0.963	.01	0	51
Software Developers, Applications & Systems Software	2703	0.961	.0281	2.04	49
Postsecondary Teachers	2655	0.943	.246	15.7	51
Maids and Housekeeping Cleaners	2643	0.939	.0114	0	51
First-Line Supervisors of Office & Admin Support Workers	2588	0.920	.0781	1.96	51
...	...	...	...	...	
...	...	...	...	...	
...	...	...	...	...	
Tool Grinders, Filers, and Sharpeners	10	0.004	.1	0	4
Average	641	.2278	.17850	18.52	27.59

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Table 11: Current (end of year) State of Residence Frequency Table

State	Frequency	Total	Percent
California	24022	281404	8.536
Texas	15837	281404	5.628
Florida	11712	281404	4.162
New York	11182	281404	3.974
Illinois	8039	281404	2.857
Pennsylvania	7837	281404	2.785
Ohio	7367	281404	2.618
Massachusetts	6458	281404	2.295
North Carolina	6348	281404	2.256
Michigan	6259	281404	2.224
Georgia	6193	281404	2.201
Louisiana	6157	281404	2.188
Montana	5537	281404	1.968
New Jersey	5421	281404	1.926
Washington	5331	281404	1.894
West Virginia	5268	281404	1.872
Tennessee	5267	281404	1.872
District of Columbia	5257	281404	1.868
Alabama	5255	281404	1.867
Virginia	5217	281404	1.854
Indiana	4809	281404	1.709
Arkansas	4804	281404	1.707
Utah	4626	281404	1.644
Mississippi	4569	281404	1.624
Idaho	4516	281404	1.605
North Dakota	4452	281404	1.582
New Mexico	4382	281404	1.557
Oregon	4367	281404	1.552
Arizona	4352	281404	1.547
South Carolina	4274	281404	1.519
New Hampshire	4236	281404	1.505
Oklahoma	4205	281404	1.494
Wyoming	4197	281404	1.491
Wisconsin	4195	281404	1.491
Minnesota	4115	281404	1.462
Missouri	3987	281404	1.417
Hawaii	3964	281404	1.409
Vermont	3946	281404	1.402
Nebraska	3927	281404	1.396
Maryland	3866	281404	1.374
Colorado	3747	281404	1.332
Kansas	3692	281404	1.312
Iowa	3579	281404	1.272
Nevada	3542	281404	1.259
South Dakota	3454	281404	1.227
Alaska	3392	281404	1.205
Kentucky	3208	281404	1.140
Delaware	3056	281404	1.086
Connecticut	3049	281404	1.083
Rhode Island	2554	281404	0.908
Maine	2378	281404	0.845

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

8.1.1 Wages Regression: k-means identified licensed individuals have higher incomes

Table 12: Affect of Occupational Licensing on Personal Income, 2016-2020

	(1) Total Income (in 1000s of $)	(2) Total Income (in 1000s of $)	(3) Total Income (in 1000s of $)
Licensed (k-means corrrected with 2 clusters)	2.0531***		
	(0.5371)		
Licensed (k-means corrrected with 3 clusters)		4.0999***	
		(0.9941)	
Licensed (self-reported)			5.5436***
			(0.3925)
Age	1.4230***	1.4286***	1.4024***
	(0.0612)	(0.0674)	(0.0612)
Age Squared	-0.0096***	-0.0097***	-0.0094***
	(0.0007)	(0.0007)	(0.0007)
Gender: Male	18.9556***	18.4024***	18.8598***
	(0.3081)	(0.3381)	(0.3081)
Race: White	3.1871***	2.9239***	3.0440***
	(0.4814)	(0.5299)	(0.4814)
Race: Black	-1.0071	-1.2228*	-1.0839*
	(0.6316)	(0.7029)	(0.6314)
Educ: High School completed or equivalent	5.5674***	5.3032***	5.4773***
	(0.5305)	(0.5666)	(0.5304)
Educ: Some college, Associate or Bachelor Degree	14.5545***	14.3116***	14.3000***
	(0.5300)	(0.5674)	(0.5301)
Educ: Has a Master Degree, Professional or Doctorate Degree	39.5273***	39.3122***	38.8812***
	(0.6757)	(0.7477)	(0.6771)
Marst: Never married/single	-5.4201***	-5.5949***	-5.2928***
	(0.3821)	(0.4250)	(0.3821)
Marst: Separated	-2.1338***	-2.3715***	-2.1442***
	(0.3733)	(0.4177)	(0.3732)
# of own children in household	1.5621***	1.6081***	1.5571***
	(0.1288)	(0.1443)	(0.1287)
Home: Owned (or being bought)	5.8405***	5.7878***	5.7523***
	(0.2979)	(0.3291)	(0.2979)
Regression	OLS	OLS	OLS
Controls	Yes 7	Yes 7	Yes 7
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r2	.2526355	.2646967	.2531305
N	281,382	224,695	281,382

Dependent variable personal income (in 1000s of dollars). Constant term is not reported.
* $p < .10$, ** $p < .05$, *** $p < .01$

8.1.2 Employment Status Regression: k-means identified licensed individuals likely to be employed

Table 13: Affect of Occupational Licensing on Employment Status, 2016-2020

	(1) Unemployed Worker	(2) Unemployed Worker	(3) Unemployed Worker
Licensed (k-means corrrected with 2 clusters)	-0.0054***		
	(0.0015)		
Licensed (k-means corrrected with 3 clusters)		-0.0112***	
		(0.0029)	
Licensed (self-reported)			-0.0111***
			(0.0011)
Age	-0.0009***	-0.0009***	-0.0008***
	(0.0002)	(0.0002)	(0.0002)
Age Squared	0.0000***	0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)
Gender: Male	0.0016*	0.0013	0.0018**
	(0.0009)	(0.0010)	(0.0009)
Race: White	-0.0080***	-0.0085***	-0.0078***
	(0.0014)	(0.0016)	(0.0014)
Race: Black	0.0175***	0.0214***	0.0177***
	(0.0018)	(0.0021)	(0.0018)
Educ: High School completed or equivalent	-0.0107***	-0.0109***	-0.0105***
	(0.0015)	(0.0017)	(0.0015)
Educ: Some college, Associate or Bachelor Degree	-0.0151***	-0.0149***	-0.0146***
	(0.0015)	(0.0017)	(0.0015)
Educ: Has a Master Degree, Professional or Doctorate Degree	-0.0125***	-0.0121***	-0.0114***
	(0.0020)	(0.0022)	(0.0020)
Marst: Never married/single	0.0194***	0.0199***	0.0191***
	(0.0011)	(0.0013)	(0.0011)
Marst: Separated	0.0146***	0.0144***	0.0146***
	(0.0011)	(0.0012)	(0.0011)
# of own children in household	0.0015***	0.0018***	0.0016***
	(0.0004)	(0.0004)	(0.0004)
Home: Owned (or being bought)	-0.0089***	-0.0095***	-0.0087***
	(0.0009)	(0.0010)	(0.0009)
Total Personal Income (in 1000s)	-0.0001***	-0.0001***	-0.0001***
	(0.0000)	(0.0000)	(0.0000)
Regression	OLS	OLS	OLS
Controls	Yes 8	Yes 8	Yes 8
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r2	.0431333	.0470275	.0434213
N	281,382	224,695	281,382

Dependent variable is a dummy measuring if a person is unemployed (experienced worker), as opposed to employed/at work or has job (but not at work last week). Constant term is not reported.
* $p < .10$, ** $p < .05$, *** $p < .01$

8.1.3 Log number of people in state licensed occupation is lower than in unlicensed occupations, suggesting regulatory capture effect dominates adverse selection effect

Table 14: Affect of Occupational Licensing on Number of People within an Occupation-State-Time cluster, 2016-2020

	(1) Log of # of people in State-Occupation-Time cluster	(2) Log of # of people in State-Occupation-Time cluster	(3) Log of # of people in State-Occupation-Time cluster
Licensed occ-state-time (k-means corrrected with 2 clusters)	-0.0548***		
	(0.0076)		
Licensed occ-state-time (k-means corrrected with 3 clusters)		-0.1466***	
		(0.0105)	
Licensed individuals in occ-state-time cluster (self-reported)			-0.0252***
			(0.0097)
Average age in an occ-state-time cluster	-0.0003	-0.0003	-0.0003
	(0.0003)	(0.0003)	(0.0003)
Proprtion of Male in occ-state-time cluster	0.0029	-0.0022	0.0025
	(0.0080)	(0.0084)	(0.0080)
Proportion of White in occ-state-time cluster	-0.0425***	-0.0401***	-0.0435***
	(0.0127)	(0.0133)	(0.0127)
Proportion of Black in occ-state-time cluster	-0.0536***	-0.0539***	-0.0538***
	(0.0166)	(0.0174)	(0.0166)
Proprtion with high school equi. educ in occ-state-time cluster	-0.0206	-0.0234	-0.0213
	(0.0142)	(0.0146)	(0.0142)
Proprtion with some high school educ in occ-state-time cluster	-0.0431***	-0.0457***	-0.0456***
	(0.0141)	(0.0146)	(0.0141)
Proprtion with advanced degrees an occ-state-time cluster	0.0026	0.0018	-0.0013
	(0.0179)	(0.0187)	(0.0179)
Proprtion of single inds. in occ-state-time cluster	0.0100	0.0057	0.0107
	(0.0095)	(0.0099)	(0.0095)
Proprtion of separated inds. in left in occ-state-time cluster	0.0026	-0.0005	0.0025
	(0.0094)	(0.0099)	(0.0094)
Average number of children in an occ-state-time cluster	0.0010	-0.0002	0.0008
	(0.0031)	(0.0033)	(0.0031)
Proportion of inds owning their home	-0.0097	-0.0125	-0.0100
	(0.0077)	(0.0080)	(0.0077)
Average personal income an occ-state-time cluster	0.0002***	0.0002***	0.0002***
	(0.0001)	(0.0001)	(0.0001)
Regression	OLS	OLS	OLS
Controls	Yes 8	Yes 8	Yes 8
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r2	.702799	.713909	.7025603
N	58,093	48,875	58,093

Dependent variable (migration) is a dummy. Constant term is not reported.
* $p < .10$, ** $p < .05$, *** $p < .01$

8.1.4 In-Migration Specification likely to be endogenous due to self-selection

Table 15: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Migrated: Moved from a different state	(2) Migrated: Moved from abroad / different nation	(3) Migrated: Moved from a different county within state	(4) Migrated: Moved within county in the same state	(5) Migrated: No / Stayed in Same House (Non-Mover)
Govt. issued prof-cert required for job / Licensed	-.001294*	-.001193***	.0006325	.0008053	.001049
	(.0007574)	(.0002713)	(.000883)	(.001423)	(.001759)
Home: Owned (or being bought)	-.02982***	-.005779***	-.02976***	-.0904***	.1558***
	(.0007532)	(.0003106)	(.0008264)	(.00134)	(.001651)
Age	-.001579***	-.0002331***	-.001769***	-.005628***	.00921***
	(.0001271)	(.00004985)	(.0001425)	(.0002339)	(.0002852)
Age Squared	.00001194***	1.566e-06***	.00001382***	.00004193***	-.00006925***
	(1.285e-06)	(4.935e-07)	(1.447e-06)	(2.371e-06)	(2.902e-06)
Gender: Male	7.279e-06	.001067***	-.0002538	-.0007212	-.00009887
	(.0006706)	(.0002795)	(.0007359)	(.001223)	(.001499)
Race: White	-.001397	-.006304***	.00185	.0004544	.005396**
	(.001123)	(.0006768)	(.001135)	(.001994)	(.002453)
Race: Black	-.005225***	-.006293***	-.003045**	-.00433	.01889***
	(.001471)	(.0008013)	(.001535)	(.002651)	(.003256)
Educ: High School completed or equivalent	.005676***	.000414	.006497***	.01552***	-.02811***
	(.0009691)	(.0005206)	(.001228)	(.002162)	(.002575)
Educ: Some college, Associate or Bachelor Degree	.01001***	.001134**	.009983***	.0145***	-.03563***
	(.0009976)	(.0005175)	(.001244)	(.002142)	(.002573)
Educ: Has a Master Degree, Professional or Doctorate Degree	.01608***	.003688***	.009802***	.01795***	-.04752***
	(.001412)	(.0007042)	(.001551)	(.002639)	(.003242)
Marst: Never married/single	-.001894**	-.001154***	.00126	.0003166	.001471
	(.0008647)	(.0003591)	(.0009362)	(.00157)	(.001921)
Marst: Separated	.00228***	.0005254*	.005648***	.01649***	-.02495***
	(.0007475)	(.000304)	(.0008645)	(.001415)	(.001744)
# of own children under 5 in household	-.003436***	-.0007213***	-.0009418	.004514***	.0005857
	(.000651)	(.0002796)	(.0007549)	(.001326)	(.001581)
Total Personal Income (in 1000s)	-3.656e-06	-8.112e-06***	7.704e-06*	.0000103*	-6.241e-06
	(4.128e-06)	(1.486e-06)	(4.119e-06)	(6.143e-06)	(8.075e-06)
Regression	OLS	OLS	OLS	OLS	OLS
Controls	Yes (8)	Yes (8)	Yes (8)	Yes (8)	Yes (8)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes	Yes
Interacted Occupation-Time FE	Yes	Yes	Yes	Yes	Yes
Interacted Occupation-State FE	Yes	Yes	Yes	Yes	Yes
Interacted State-Time FE	Yes	Yes	Yes	Yes	Yes
r2	.08924	.06965	.07971	.1129	.1515
N	278500	278500	278500	278500	278500

Dependent variable (migration) is a dummy.

* $p < .10$, ** $p < .05$, *** $p < .01$

8.2 Migration to Another State (Transformed Data)

Table 16: Affect of Occupational Licensing on Migration, 2016-2019

	(1)	(2)	(3)	(4)	(5)	(6)
Licensed (k-means corrected with 2 clusters)	0.0005	0.0007	0.0007	0.0007	0.0006	0.0007
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)
Proportion Licensed (prob. of re-licensing costs)	-0.0017	0.0000	0.0001	0.0001	-0.0043***	-0.0039**
	(0.0015)	(0.0015)	(0.0015)	(0.0015)	(0.0016)	(0.0016)
Age		-0.0016***	-0.0016***	-0.0016***	-0.0018***	-0.0012***
		(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Age Squared		0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Gender: Male			0.0001	0.0002	0.0005	0.0005
			(0.0006)	(0.0006)	(0.0006)	(0.0006)
Race: White				-0.0027**	-0.0019*	-0.0020*
				(0.0011)	(0.0011)	(0.0011)
Race: Black				-0.0005	0.0008	0.0006
				(0.0015)	(0.0015)	(0.0015)
Educ: High School completed or equivalent					0.0069***	0.0056***
					(0.0010)	(0.0010)
Educ: Some college, Associate or Bachelor Degree					0.0114***	0.0101***
					(0.0010)	(0.0010)
Educ: Masters, Professional or Doctorate Degree					0.0198***	0.0186***
					(0.0013)	(0.0013)
# of own children in household						-0.0037***
						(0.0003)
Regression	OLS	OLS	OLS	OLS	OLS	OLS
Controls	No	Yes (1)	Yes (2)	Yes (3)	Yes (4)	Yes (5)
Time FE	No	No	No	No	No	No
State FE	No	No	No	No	No	No
Occupation FE	No	No	No	No	No	No
r2	9.38e-06	.0054968	.005497	.0055497	.0068227	.0077132
N	200,437	200,437	200,437	200,437	200,437	200,437

Dependant variable is binary migration variable measuring if a person moved to another state over the next one year.
* $p < .10$, ** $p < .05$, *** $p < .01$

Table 17: Affect of Occupational Licensing on Migration, 2016-2019

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Licensed (k-means corrected with 2 clusters)	0.0007	0.0011	0.0007	0.0008	0.0007	0.0011	0.0011	0.0008	0.0008
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)
Proportion Licensed (prob. of re-licensing costs)	-0.0039**	-0.0040**	-0.0036	-0.0037**	0.0000	-0.0037	-0.0041	-0.0034	0.0000
	(0.0016)	(0.0016)	(0.0068)	(0.0016)	(.)	(0.0068)	(0.0068)	(0.0068)	(.)
Age	-0.0012***	-0.0011***	-0.0013***	-0.0011***	-0.0013***	-0.0013***	-0.0013***	-0.0013***	-0.0013**
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Age Squared	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000**
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Gender: Male	0.0005	0.0004	-0.0009	0.0004	-0.0009	-0.0010	-0.0010	-0.0010	-0.0011
	(0.0006)	(0.0006)	(0.0007)	(0.0006)	(0.0007)	(0.0007)	(0.0007)	(0.0007)	(0.0007)
Race: White	-0.0020*	-0.0020	-0.0017	-0.0019	-0.0017	-0.0017	-0.0017	-0.0016	-0.0017
	(0.0011)	(0.0012)	(0.0011)	(0.0012)	(0.0011)	(0.0013)	(0.0013)	(0.0012)	(0.0013)
Race: Black	0.0006	0.0001	0.0018	0.0001	0.0017	0.0014	0.0014	0.0015	0.0014
	(0.0015)	(0.0016)	(0.0015)	(0.0016)	(0.0015)	(0.0016)	(0.0016)	(0.0016)	(0.0016)
Educ: High School completed or equivalent	0.0056***	0.0060***	0.0052***	0.0060***	0.0053***	0.0056***	0.0056***	0.0056***	0.0057**
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)
Educ: Some college, Associate or Bachelor Degree	0.0101***	0.0101***	0.0083***	0.0101***	0.0084***	0.0085***	0.0085***	0.0085***	0.0085**
	(0.0010)	(0.0010)	(0.0011)	(0.0010)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)
Educ: Masters, Professional or Doctorate Degree	0.0186***	0.0181***	0.0134***	0.0181***	0.0136***	0.0132***	0.0132***	0.0131***	0.0132**
	(0.0013)	(0.0013)	(0.0015)	(0.0013)	(0.0015)	(0.0015)	(0.0015)	(0.0015)	(0.0015)
# of own children in household	-0.0037***	-0.0038***	-0.0036***	-0.0038***	-0.0036***	-0.0037***	-0.0037***	-0.0037***	-0.0037**
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Regression	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)	Yes (5)	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes
State FE	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	Yes	No	No	No	Yes	Yes
Interacted Occupation-Time FE	No	No	No	No	Yes	No	No	No	Yes
r2	.0077257	.0114298	.0110022	.0135405	.0166205	.0147039	.0147205	.0168103	.022447
N	200,437	200,437	200,429	200,437	200,357	200,429	200,429	200,429	200,357

Dependant variable is binary migration variable measuring if a person moved to another state over the next one year.

* $p < .10$, ** $p < .05$, *** $p < .01$

8.2.1 Out-Migration Specification with Full Set of Controls does not change results

Table 18: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0012	0.0009	-0.0010	-0.0011
	(0.0012)	(0.0014)	(0.0022)	(0.0027)
Proportion Licensed (prob. of re-licensing costs)	-0.0038	-0.0050	0.0062	0.0027
	(0.0068)	(0.0079)	(0.0126)	(0.0155)
Home: Owned (or being bought)	-0.0283***	-0.0286***	-0.0879***	0.1448***
	(0.0009)	(0.0009)	(0.0015)	(0.0019)
Marst: Never married/single	-0.0035***	-0.0011	-0.0022	0.0068***
	(0.0010)	(0.0011)	(0.0018)	(0.0022)
Marst: Separated	0.0021**	0.0040***	0.0150***	-0.0210***
	(0.0008)	(0.0010)	(0.0016)	(0.0019)
Total Personal Income (in 1000s)	-0.0000	0.0000**	0.0000**	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Age	-0.0013***	-0.0016***	-0.0055***	0.0084***
	(0.0001)	(0.0002)	(0.0003)	(0.0003)
Age Squared	0.0000***	0.0000***	0.0000***	-0.0001***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Gender: Male	-0.0005	-0.0007	-0.0003	0.0015
	(0.0007)	(0.0008)	(0.0014)	(0.0017)
Race: White	-0.0004	0.0001	0.0008	-0.0005
	(0.0012)	(0.0013)	(0.0022)	(0.0026)
Race: Black	-0.0025	-0.0035**	-0.0026	0.0086**
	(0.0016)	(0.0017)	(0.0029)	(0.0036)
Educ: High School completed or equivalent	0.0065***	0.0055***	0.0192***	-0.0311***
	(0.0011)	(0.0014)	(0.0024)	(0.0028)
Educ: Some college, Associate or Bachelor Degree	0.0101***	0.0090***	0.0182***	-0.0373***
	(0.0011)	(0.0014)	(0.0024)	(0.0028)
Educ: Masters, Professional or Doctorate Degree	0.0153***	0.0089***	0.0232***	-0.0474***
	(0.0015)	(0.0017)	(0.0029)	(0.0035)
# of own children in household	-0.0032***	-0.0019***	-0.0026***	0.0078***
	(0.0003)	(0.0003)	(0.0005)	(0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (8)	Yes (8)	Yes (8)	Yes (8)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	No
Interacted Occupation-Time FE	No	No	No	No
r2	.0234229	.0187814	.056046	.0889917
N	200,429	200,429	200,429	200,429

Dependent variable (migration) is a dummy.
* $p < .10$, ** $p < .05$, *** $p < .01$

8.2.2 Out-Migration Specification limited to continuing workers does not change results

Table 19: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0013	0.0007	-0.0014	-0.0007
	(0.0012)	(0.0014)	(0.0024)	(0.0029)
Proportion Licensed (prob. of re-licensing costs)	-0.0022	-0.0118	0.0106	0.0033
	(0.0067)	(0.0081)	(0.0135)	(0.0165)
Age	-0.0010***	-0.0014***	-0.0057***	0.0082***
	(0.0001)	(0.0002)	(0.0003)	(0.0003)
Age Squared	0.0000***	0.0000***	0.0000***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Gender: Male	-0.0009	-0.0005	-0.0011	0.0025
	(0.0007)	(0.0008)	(0.0015)	(0.0018)
Race: White	-0.0013	-0.0016	-0.0034	0.0063**
	(0.0012)	(0.0013)	(0.0024)	(0.0028)
Race: Black	0.0009	-0.0026	0.0131***	-0.0114***
	(0.0016)	(0.0017)	(0.0032)	(0.0038)
Educ: High School completed or equivalent	0.0050***	0.0046***	0.0161***	-0.0258***
	(0.0010)	(0.0014)	(0.0026)	(0.0031)
Educ: Some college, Associate or Bachelor Degree	0.0068***	0.0072***	0.0116***	-0.0256***
	(0.0010)	(0.0014)	(0.0026)	(0.0031)
Educ: Masters, Professional or Doctorate Degree	0.0102***	0.0069***	0.0156***	-0.0327***
	(0.0014)	(0.0017)	(0.0031)	(0.0038)
# of own children in household	-0.0033***	-0.0026***	-0.0058***	0.0117***
	(0.0002)	(0.0003)	(0.0005)	(0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
r2	.0134225	.0107423	.0322574	.0446523
N	175,659	175,659	175,659	175,659

Dependent variable (migration) is a dummy.
* $p < .10$, ** $p < .05$, *** $p < .01$

8.3 k-means with 3 clusters

Table 20: Summary Statistics: kmean 3 cluster drops "Not Sure category".

	Mean	Std. Dev.	Min	Max	Universe
Licensed (Self-Reported)	0.158	0.30	0	1	157862
Licensed at beginning of year (k-means corrected with 3 clusters)	0.150	0.36	0	1	157862
Proportion of people licensed in an occupation (3 kmean)	0.150	0.32	0	1	157862
Migrated: Moved to different state	0.018	0.13	0	1	157862
Migrated: Moved to abroad / different nation	0.000	0.00	0	0	157862
Migrated: Moved to different county within state	0.022	0.15	0	1	157862
Migrated: Moved to different house within county	0.066	0.25	0	1	157862
Migrated: No / Stayed in Same House (Non-Mover)	0.895	0.31	0	1	157862
Home: Owned (or being bought)	0.688	0.46	0	1	157862
Home: Rented	0.312	0.46	0	1	157862
Age	43.085	14.41	16	79	157862
Gender: Male	0.515	0.50	0	1	157862
Gender: Female	0.485	0.50	0	1	157862
Race: White	0.817	0.39	0	1	157862
Race: Black	0.092	0.29	0	1	157862
Race: Other	0.091	0.29	0	1	157862
Educ: Has some schooling	0.079	0.27	0	1	157862
Educ: High School completed or equivalent	0.265	0.44	0	1	157862
Educ: Some college, Associate or Bachelor Degree	0.523	0.50	0	1	157862
Educ: Has a Master Degree, Professional or Doctorate Degree	0.134	0.34	0	1	157862
Marst: Married	0.543	0.50	0	1	157862
Marst: Never married/single	0.298	0.46	0	1	157862
Marst: Separated	0.159	0.37	0	1	157862
# of own children in household	0.802	1.12	0	9	157862
Total Personal Income (in 1000s)	61.996	79.92	-13	2085.7	157862
State of Residence at beginning of year			2	52	157862
Occupation at beginning of year			1	443	157862
Beginning of Year (March, 20XX)			2016	2019	157862
Observations	157862				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020

Licensed (self-reported) binary variable takes on a value of 1 if the individual requires a govt. issued professional certificate for their primary job, and 0 otherwise. It was created from the three certification questions.

Table 21: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 3 clusters)	0.0033	-0.0018	-0.0055	0.0040
	(0.0023)	(0.0024)	(0.0039)	(0.0049)
Proportion Licensed (prob. of re-licensing costs)	0.0073	0.0008	0.0085	-0.0166
	(0.0092)	(0.0104)	(0.0167)	(0.0208)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	No
Interacted Occupation-Time FE	No	No	No	No
r2	.0151968	.011955	.0315694	.0458892
N	157,854	157,854	157,854	157,854

Dependent variable (migration) is a dummy.
* $p < .10$, ** $p < .05$, *** $p < .01$
Observations are fewer than 200,437 due to “Not Sure” cluster values being dropped.
Individual level controls such as age, age squared, gender, race, education and number of children are not reported. Constant term is not reported.